

Mass Spectra and Decay Widths of Heavy Mesons in the Gaussian Expansion Method

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Heavy mesons that carry at least one bottom or charm quark give us an excellent testing ground for the non-relativistic effective descriptions of Quantum Chromodynamics (QCD). In this work we compute the mass spectra and a set of decay widths for the bottomonium ($b\bar{b}$), charmonium ($c\bar{c}$), B_c ($b\bar{c}$), and D ($c\bar{u}$) sectors. We do this by solving the two-body Schrödinger equation for the Cornell potential, supplemented by the perturbative Breit–Fermi spin-dependent corrections. To solve the bound-state problem we use the Gaussian Expansion Method (GEM), which turns it into a generalized algebraic eigenvalue problem in which every Hamiltonian and overlap matrix element is evaluated in closed form. We then tune the Cornell parameters to the Particle Data Group masses through a true (dimensionless) χ^2 minimization that also folds in the ground- and first-excited leptonic widths, weighting each residual by a physically motivated uncertainty in which a 10 MeV theory systematic dominates the negligible experimental mass errors. The fit reproduces the bottomonium spectrum very well, with a root-mean-square (RMS) deviation of 7.0 MeV ($\chi^2/\text{dof} = 0.67$), and the charmonium spectrum with 28.8 MeV ($\chi^2/\text{dof} = 10.3$); the larger charm deviations sit, as expected, in the relativistic D -wave and the high radial excitations. A leave-one-out cross-validation confirms that our rigid four-parameter form generalizes rather than overfits. Using the same wavefunctions we also evaluate the leptonic, two-photon, and radiative ($E1/M1$) transition widths together with a 3P_0 open-flavour width. The $E1$ and $M1$ transitions agree with experiment at the 10–25% level and the ground-state leptonic widths within $\sim 15\%$, whereas the two-photon width of η_c comes out too large by a factor of ~ 2.4 . We trace this last discrepancy quantitatively to the non-relativistic over-concentration of the spin-singlet wavefunction at the origin, and not to the variational method itself.

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I. INTRODUCTION AND THEORETICAL OVERVIEW

Quantum Chromodynamics (QCD) is asymptotically free at short distances and confining at long distances. The latter regime is non-perturbative, and no closed-form interquark potential can be derived from first principles. Heavy mesons—bound states of a heavy quark and antiquark, such as charmonium ($c\bar{c}$) and bottomonium ($b\bar{b}$)—are nonetheless amenable to a controlled approximation, because the constituent mass m_Q is large compared with the confinement scale $\Lambda_{\text{QCD}} \approx 0.2$ GeV. The relevant degrees of freedom then separate into a hierarchy of scales: the hard scale m_Q , the soft scale $m_Q v$ of the relative momentum, and the ultrasoft scale $m_Q v^2$ of the binding energy, where v is the typical quark velocity. The non-relativistic expansion parameter is v^2 , with $v^2 \approx 0.1$ for bottomonium and $v^2 \approx 0.3$ for charmonium. This scale separation is the basis of non-relativistic effective theories of QCD (NRQCD and pNRQCD) [1, 2] and justifies treating heavy mesons with a non-relativistic Schrödinger equation: a fully relativistic Bethe–Salpeter or Dirac treatment is not required, while relativistic spin-dependent corrections enter perturbatively at order $\mathcal{O}(v^2/c^2)$ and are responsible for the fine and hyperfine structure of the spectrum.

Rather than a lattice or effective-field-theory calculation, we adopt the phenomenological Cornell potential [3, 4], which combines a short-range Coulombic term with a long-range linear confining term, augmented by the Breit–Fermi spin-dependent corrections [5, 6]. Constituent quark models of this type have a long record of success in heavy-meson spectroscopy [7–9]. Because this potential is not solvable in closed form, we expand the wavefunction in a non-orthogonal Gaussian basis and apply the Rayleigh–Ritz variational principle. This combination is the Gaussian Expansion Method (GEM) [10, 11], a few-body variational technique here applied to the two-body sector. The GEM has already been used for heavy-quark systems: it reproduces the charmonium spectrum in a QCD-inspired quark potential model [12], and it describes the spectrum of singly and doubly heavy baryons in the quark model [13]. This record for both mesonic and baryonic heavy-quark states is what motivates us to carry the method over to the mixed B_c and heavy-light D sectors as well. The resulting matrix elements are all analytic, so the continuum eigenvalue problem reduces to a finite generalized algebraic eigenvalue problem.

We also treat the open-flavour B_c ($b\bar{c}$) and D ($c\bar{u}$) sectors. The B_c system is the only meson built from two heavy quarks of different flavour, and it has been studied extensively in the constituent quark model [14]. The

D meson, on the other hand, is a heavy-light system in which the light quark is not slow, so the non-relativistic treatment is the least justified there; we still report those results for completeness, but with this caveat kept in mind.

Three things set our implementation apart from a textbook Cornell calculation. First, we reduce every required matrix element—including the singular $1/r^3$ spin-orbit and tensor integrals—to a closed Gamma function, so there is no numerical quadrature and no grid artefact at all. Second, our parameter fit is a genuine statistical procedure: we normalize the residuals by a physical error model in which a theory systematic, not the sub-MeV experimental error, sets the scale, and we then validate the fit against overfitting by a leave-one-out cross-validation (Sec. III). Third, we read all of the reference data—the masses and the leptonic and two-photon partial widths—directly from the Particle Data Group (PDG) database [15], so that the whole analysis stays reproducible from a single source.

A. Hamiltonian for Heavy Mesons

The two-body problem of a quark of mass m_1 and an antiquark of mass m_2 separated by $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ reduces, after removal of the center-of-mass motion, to an effective one-body problem with reduced mass $\mu = m_1 m_2 / (m_1 + m_2)$. The radial kinetic operator is

$$T = -\frac{1}{2\mu} \nabla^2, \quad (1)$$

which, acting on a state of orbital angular momentum l , contributes both a radial part and the centrifugal barrier $l(l+1)/2\mu r^2$. Throughout we work in natural units, $\hbar = c = 1$. The full Hamiltonian is

$$H = T + V_{\text{Cornell}} + V_{LS} + V_{SS} + V_T, \quad (2)$$

where V_{Cornell} is the spin-independent potential and V_{LS} , V_{SS} , and V_T are the spin-orbit, spin-spin, and tensor Breit–Fermi corrections. The bound-state mass of a state α is $M_\alpha = m_1 + m_2 + E_\alpha$, with E_α the corresponding eigenvalue of Eq. (2).

1. Cornell Potential

The Cornell potential models one-gluon exchange (OGE) at short distance and the confining QCD string at long distance:

$$V_{\text{Cornell}}(r) = -\frac{4}{3} \frac{\alpha_s}{r} + b r + c. \quad (3)$$

The colour factor $4/3$ is the Casimir invariant $C_F = (N_c^2 - 1)/2N_c$ with $N_c = 3$, α_s is the strong coupling (scale-dependent at short distances), b is the string tension, and c is an overall constant shift. In this work α_s , b , and c are treated as effective parameters fixed by a fit to data (Sec. II B).

2. Spin-Orbit Coupling

The spin-dependent Breit–Fermi corrections follow from the non-relativistic reduction of the QCD interaction kernel [4, 7] and depend explicitly on the Lorentz transformation properties of each part of the potential. One-gluon exchange is mediated by a spin-1 gluon and therefore behaves as a Lorentz *vector*; the linear confining term, which models the QCD flux tube, is universally taken to be a Lorentz *scalar* [6]. A vector interaction produces a positive spin-orbit contribution, whereas for a scalar interaction the relativistic Thomas precession reverses the sign. The spin-orbit potential for a general unequal-mass system is therefore

$$V_{LS}(r) = \mathbf{L} \cdot \mathbf{S} \left[\frac{4\alpha_s}{3} \left(\frac{1}{m_1 m_2} + \frac{1}{4m_1^2} + \frac{1}{4m_2^2} \right) \frac{1}{r^3} - \frac{b}{4} \left(\frac{1}{m_1^2} + \frac{1}{m_2^2} \right) \frac{1}{r} \right], \quad (4)$$

with $\langle \mathbf{L} \cdot \mathbf{S} \rangle = \frac{1}{2}[J(J+1) - L(L+1) - S(S+1)]$. The first term is the positive vector (OGE) spin-orbit interaction; the negative sign of the second term is the Thomas-precession contribution of the scalar confining potential. For equal masses ($m_1 = m_2 = m$) Eq. (4) reduces to the familiar $(2\alpha_s/m^2)r^{-3} - (b/2m^2)r^{-1}$ form.

For the unequal-mass B_c system the spin operator must in principle be decomposed into symmetric and antisymmetric parts, $\mathbf{S}_\pm = \mathbf{S}_1 \pm \mathbf{S}_2$. The antisymmetric $\mathbf{L} \cdot \mathbf{S}_-$ term breaks the C -parity classification and mixes states of equal J but different total spin (e.g. the 1P_1 and 3P_1 levels). The B_c masses that we compare with data are the $L = 0$ states, for which the spin-orbit interaction vanishes identically; we report the full B_c P - and D -wave multiplet as predictions, but the residual $\mathbf{S}_-$ mixing of those levels is left for future work.

3. Spin-Spin Coupling

The hyperfine spin-spin interaction splits the pseudoscalar and vector ground states (e.g. η_c and J/ψ). It originates from the chromomagnetic part of one-gluon exchange [4, 6] and is a contact interaction,

$$V_{SS}(r) = \frac{32\pi\alpha_s}{9m_1m_2} \mathbf{S}_1 \cdot \mathbf{S}_2 \delta^{(3)}(\mathbf{r}), \quad (5)$$

with $\langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle = \frac{1}{2}[S(S+1) - \frac{3}{2}]$, equal to $-3/4$ for spin singlets and $+1/4$ for spin triplets. A point-like $\delta^{(3)}(\mathbf{r})$ in three dimensions is too singular for a non-relativistic Schrödinger operator—it has no normalizable ground state—so it must be regularized. Following the relativized quark model, we replace it by a normalized Gaussian of width σ [7],

$$\delta^{(3)}(\mathbf{r}) \longrightarrow \tilde{\delta}^{(3)}(\mathbf{r}) = \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 e^{-\sigma^2 r^2}. \quad (6)$$

The smearing scale σ is a genuine physical parameter: it acts as a short-distance cutoff and represents the finite extent over which the contact term is resolved in the effective theory. Because the hyperfine splitting depends on σ , it is varied in the fit alongside the Cornell parameters and is reported together with them in Appendix B.

4. Tensor Interaction

The tensor interaction acts only on spin-triplet states with $L \geq 1$ and, in particular, mixes the 3S_1 and 3D_1 channels, so that the physical 1^{--} states are not pure L eigenstates. The tensor operator likewise descends from the spin-dependent part of one-gluon exchange and is conventionally written [4, 6, 7]

$$V_T(r) = \frac{4\alpha_s}{3m_1m_2} \frac{1}{r^3} S_{12}, \quad S_{12} = 3(\mathbf{S}_1 \cdot \hat{\mathbf{r}})(\mathbf{S}_2 \cdot \hat{\mathbf{r}}) - \mathbf{S}_1 \cdot \mathbf{S}_2. \quad (7)$$

The expectation value $\langle S_{12} \rangle$ vanishes identically for S -waves; for spin-triplet states with $L \geq 1$ its diagonal value is

$$\langle S_{12} \rangle = \begin{cases} -(L+1)/(2L-1), & J = L-1, \\ +1, & J = L, \\ -L/(2L+3), & J = L+1, \end{cases} \quad (8)$$

which gives -2 , $+1$, and $-1/5$ for the 3P_0 , 3P_1 , and 3P_2 levels, respectively. The off-diagonal 3S_1 – 3D_1 matrix element is proportional to $\sqrt{8}$ and is included by diagonalizing the corresponding 2×2 mass block, so that the physical 2^3S_1 and 1^3D_1 states (e.g. the $\psi(2S)$ and $\psi(3770)$) carry the correct S – D admixture.

B. Gaussian Expansion Method

1. Rayleigh–Ritz Variational Method

For a Hamiltonian H that is not analytically solvable, the Rayleigh–Ritz method expands the trial state in a fixed basis with adjustable linear coefficients. For any normalizable trial state ψ , the energy functional satisfies

$$E[\psi] = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0, \quad (9)$$

and, by the Hylleraas–Undheim–MacDonald theorem [16], the k -th eigenvalue obtained from an $n_{\max}$ -dimensional trial space is a rigorous upper bound to the k -th exact eigenvalue. The variational solution is obtained by requiring the energy to be stationary with respect to every linear coefficient C_n ,

$$\frac{\partial E[\psi]}{\partial C_n} = 0, \quad (10)$$

which, for a non-orthogonal basis, is equivalent to the generalized eigenvalue problem stated in Eq. (14). The

nonlinear parameters of the basis (the Gaussian widths) are not optimized individually; they are fixed in advance by the geometric prescription of Eq. (12), which is the defining feature of the GEM [10, 11].

2. Gaussian Basis

For each sector and each (l, S) channel we expand the spatial wavefunction of a physical state α in $n_{\max}$ Gaussians,

$$\psi_\alpha(\mathbf{r}) = \sum_{n=1}^{n_{\max}} C_n^{(\alpha)} \phi_{nlm}(\mathbf{r}), \quad (11a)$$

$$\phi_{nlm}(\mathbf{r}) = N_{nl} r^l e^{-\nu_n r^2} Y_{lm}(\hat{\mathbf{r}}), \quad (11b)$$

$$N_{nl} = \left[\frac{2^{l+2} (2\nu_n)^{l+3/2}}{\sqrt{\pi} (2l+1)!!} \right]^{1/2}. \quad (11c)$$

Although hydrogenic or harmonic-oscillator bases are common, the Gaussian basis is well suited here because a geometric distribution of widths covers, simultaneously, the short-range Coulomb region and the diffuse linear-confinement tail. The widths are chosen in geometric progression,

$$\nu_n = \frac{1}{r_n^2}, \quad r_n = r_{\min} a^{n-1}, \quad a = \left(\frac{r_{\max}}{r_{\min}} \right)^{1/(n_{\max}-1)}, \quad (12)$$

with $r_{\min} = 0.05 \text{ GeV}^{-1}$ and $r_{\max} = 15 \text{ GeV}^{-1}$, chosen to bracket the relevant length scales of all sectors; we use $n_{\max} = 25$. The geometric mesh both reduces the number of free parameters and limits the near-linear dependence of the (non-orthogonal) basis, keeping the overlap matrix sufficiently well conditioned for a stable solution of the generalized eigenvalue problem. Figure 1 illustrates how the 25 weighted Gaussians of a single channel build up the radial amplitude $u(r) = rR(r)$ of the quarkonium ground states, Fig. 2 extends the same construction to the $1S$, $2S$, and $3S$ levels, and Fig. 4 displays the geometric mesh and the physical length scales it spans.

II. CALCULATION OF THE MASS SPECTRA

A. Analytical Matrix Elements

With the basis of Eq. (11), every required radial integral is a Gaussian moment (the full derivations of all matrix elements in this section are collected in Appendix A),

$$\mathcal{I}_k(\nu) \equiv \int_0^\infty r^k e^{-\nu r^2} dr = \frac{1}{2} \frac{\Gamma(\frac{k+1}{2})}{\nu^{(k+1)/2}}. \quad (13)$$

Because the basis is non-orthogonal, the variational problem is a *generalized* eigenvalue problem,

$$\mathbf{H} \mathbf{c} = E \mathbf{N} \mathbf{c}, \quad (14)$$

where $\mathbf{N}$ is the overlap matrix,

$$N_{nn'} = N_{nl} N_{n'l} \mathcal{I}_{2l+2}(\nu_{nn'}) = N_{nl} N_{n'l} \frac{\Gamma(l + \frac{3}{2})}{2 \nu_{nn'}^{l+3/2}}, \quad (15)$$

and $\nu_{nn'} = \nu_n + \nu_{n'}$. The spin-independent matrix elements are

$$\begin{aligned} T_{nn'} &= \frac{2l+3}{\mu} \frac{\nu_n \nu_{n'}}{\nu_{nn'}} N_{nn'}, \\ V_{nn'}^{(c)} &= c N_{nn'}, \\ V_{nn'}^{(\text{Coul})} &= -\frac{4}{3} \alpha_s N_{nl} N_{n'l} \frac{\Gamma(l+1)}{2 \nu_{nn'}^{l+1}}, \\ V_{nn'}^{(\text{Lin})} &= b N_{nl} N_{n'l} \frac{\Gamma(l+2)}{2 \nu_{nn'}^{l+2}}. \end{aligned} \quad (16)$$

The compact kinetic term $T_{nn'}$ already incorporates the centrifugal barrier $l(l+1)/2\mu r^2$; no separate centrifugal matrix element is required. The Breit–Fermi matrix elements, for $l \geq 1$, are

$$\begin{aligned} V_{nn'}^{(LS)} &= \langle \mathbf{L} \cdot \mathbf{S} \rangle \left[\frac{4\alpha_s}{3} \left(\frac{1}{m_1 m_2} + \frac{1}{4m_1^2} + \frac{1}{4m_2^2} \right) \langle r^{-3} \rangle_{nn'} \right. \\ &\quad \left. - \frac{b}{4} \left(\frac{1}{m_1^2} + \frac{1}{m_2^2} \right) \langle r^{-1} \rangle_{nn'} \right], \\ V_{nn'}^{(T)} &= \frac{4\alpha_s}{3 m_1 m_2} \langle S_{12} \rangle \langle r^{-3} \rangle_{nn'}, \\ V_{nn'}^{(SS)} &= \frac{32\pi\alpha_s}{9 m_1 m_2} \langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 N_{nl} N_{n'l} \frac{\Gamma(l + \frac{3}{2})}{2 (\nu_{nn'} + \sigma^2)^{l+3/2}}, \end{aligned} \quad (17)$$

with $\langle r^{-3} \rangle_{nn'} = N_{nl} N_{n'l} \Gamma(l)/2\nu_{nn'}^l$ and $\langle r^{-1} \rangle_{nn'} = N_{nl} N_{n'l} \Gamma(l+1)/2\nu_{nn'}^{l+1}$. The $r^2 dr$ integration measure combined with the r^{2l} behaviour of the basis renders the integrand of the singular $1/r^3$ tensor and spin-orbit terms proportional to r^{2l-1} , which is integrable for all $l \geq 1$. The $1/r^3$ singularity is thus regularized analytically into a finite Gamma function—an advantage of the Gaussian basis over grid-based finite-difference solvers. The spin-orbit and tensor terms are evaluated as first-order expectation values over the converged eigenvectors rather than added to the Hamiltonian before diagonalization, which avoids the variational collapse that the unregularized $1/r^3$ potential would otherwise induce.

Before diagonalization, the Hamiltonian and overlap matrices are scaled by the basis norms, $H_{nn'} \rightarrow H_{nn'}/N_{nl} N_{n'l}$ and likewise for $\mathbf{N}$. This diagonal preconditioning improves the condition number of $\mathbf{N}$; the resulting symmetric-definite generalized eigenvalue problem is solved with a standard library routine. The eigenvalues yield the variational mass spectrum and the eigenvectors yield the Gaussian coefficients $C_n^{(\alpha)}$ that reconstruct each radial wavefunction (Fig. 3).

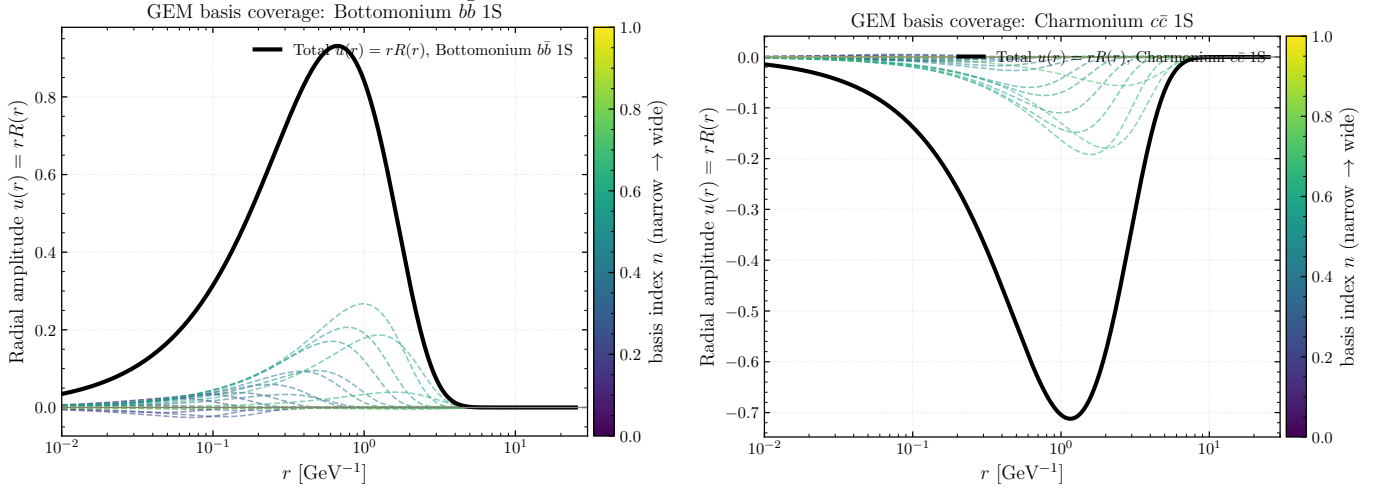


FIG. 1. Radial amplitude $u(r) = rR(r)$ of the 1S ground state in the Gaussian Expansion Method for bottomonium (left) and charmonium (right). Dashed curves are the individual weighted Gaussian basis functions on the geometric mesh of widths ν_n (narrow, left $\rightarrow$ wide, right, colour coded by basis index); the solid black curve is their converged sum. On the logarithmic r axis the narrow Gaussians build the short-range one-gluon-exchange cusp while the broad Gaussians reproduce the confinement tail; the lighter charm system (right) is visibly more diffuse.

B. χ^2 Optimization of the Cornell Parameters

Because the Cornell potential is phenomenological, its parameters are not fixed by first principles. In each sector we tune $\vec{p} = (\alpha_s, b, c, \sigma)$ by minimizing a true dimensionless χ^2 ,

$$\chi^2(\vec{p}) = \sum_i \left(\frac{M_i^{\text{calc}}(\vec{p}) - M_i^{\text{exp}}}{\sigma_i} \right)^2 + \sum_j \left(\frac{\Gamma_j^{\text{calc}}(\vec{p}) - \Gamma_j^{\text{exp}}}{\sigma_j^\Gamma} \right)^2, \quad (18)$$

against the PDG masses [15] and the measured leptonic (e^+e^-) widths of the ground- and first-excited vector states. The constituent masses m_1 and m_2 are fixed inputs and are *not* optimized. The crucial choice is the per-point uncertainty σ_i , discussed in detail in Sec. III: rather than the sub-MeV experimental error—which a non-relativistic model cannot plausibly match—we use a physically motivated $\sigma_i = \sqrt{(\sigma_i^{\text{exp}})^2 + \sigma_{\text{theory}}^2}$ with a theory floor $\sigma_{\text{theory}} = 10$ MeV on masses, and $\sigma_j^\Gamma = 0.30 \Gamma_j^{\text{exp}}$ on the leptonic widths. Including the leptonic widths in the objective ties the wavefunction at the origin to the data, so the potential is constrained by short-distance information and not by masses alone. The minimization uses a trust-region reflective least-squares algorithm, which handles parameter bounds and the exactly determined sectors (Sec. III); a plain Levenberg–Marquardt step does not. The covariance of the fitted parameters is obtained from the Jacobian at the minimum and is propagated to every predicted observable (Sec. III); the fitted parameters and their 1σ errors are listed in Appendix B.

For the open-flavour sectors the data are sparser than the parameters. The B_c sector has only two measured masses, so b and c are fitted to them while α_s

and σ are *not* free: they are fixed by a leading-order renormalization-group (log) interpolation of the $c\bar{c}$ and $b\bar{b}$ values to the B_c reduced-mass scale,

$$\frac{1}{\alpha_s^{bc}} = \frac{1}{\alpha_s^{cc}} + \frac{\ln(\mu_{bc}/\mu_{cc})}{\ln(\mu_{bb}/\mu_{cc})} \left(\frac{1}{\alpha_s^{bb}} - \frac{1}{\alpha_s^{cc}} \right), \quad (19)$$

and σ^{bc} linearly in the same variable; their uncertainties are inherited from the $c\bar{c}$ and $b\bar{b}$ fits by error propagation. The D sector is anchored by the single D^0 mass, which fixes only the offset c ; the remaining parameters are frozen to physical values ($b = 0.18$ GeV², $\alpha_s = 0.50$, $\sigma = 0.85$ GeV).

III. STATISTICAL METHODOLOGY AND UNCERTAINTY QUANTIFICATION

The statistical treatment is central to interpreting the results, so we set it out explicitly. The key point is that two conceptually different uncertainties appear, and conflating them is the commonest pitfall of potential-model fits.

A. Two distinct uncertainties

The first is the *model* uncertainty: the $\pm$ quoted on each predicted quantity. It is obtained by propagating the fitted-parameter covariance $\text{Cov}(\vec{p})$ through the (generally nonlinear) map from parameters to observable $\mathcal{O}$ by central finite differences,

$$\sigma_{\mathcal{O}}^2 = \sum_k \left(\frac{\partial \mathcal{O}}{\partial p_k} \right)^2 \sigma_{p_k}^2, \quad \frac{\partial \mathcal{O}}{\partial p_k} \approx \frac{\mathcal{O}(p_k + \epsilon) - \mathcal{O}(p_k - \epsilon)}{2\epsilon}, \quad (20)$$

GEM basis-function coverage of the radial wavefunctions (dashed: weighted Gaussians, solid: total)

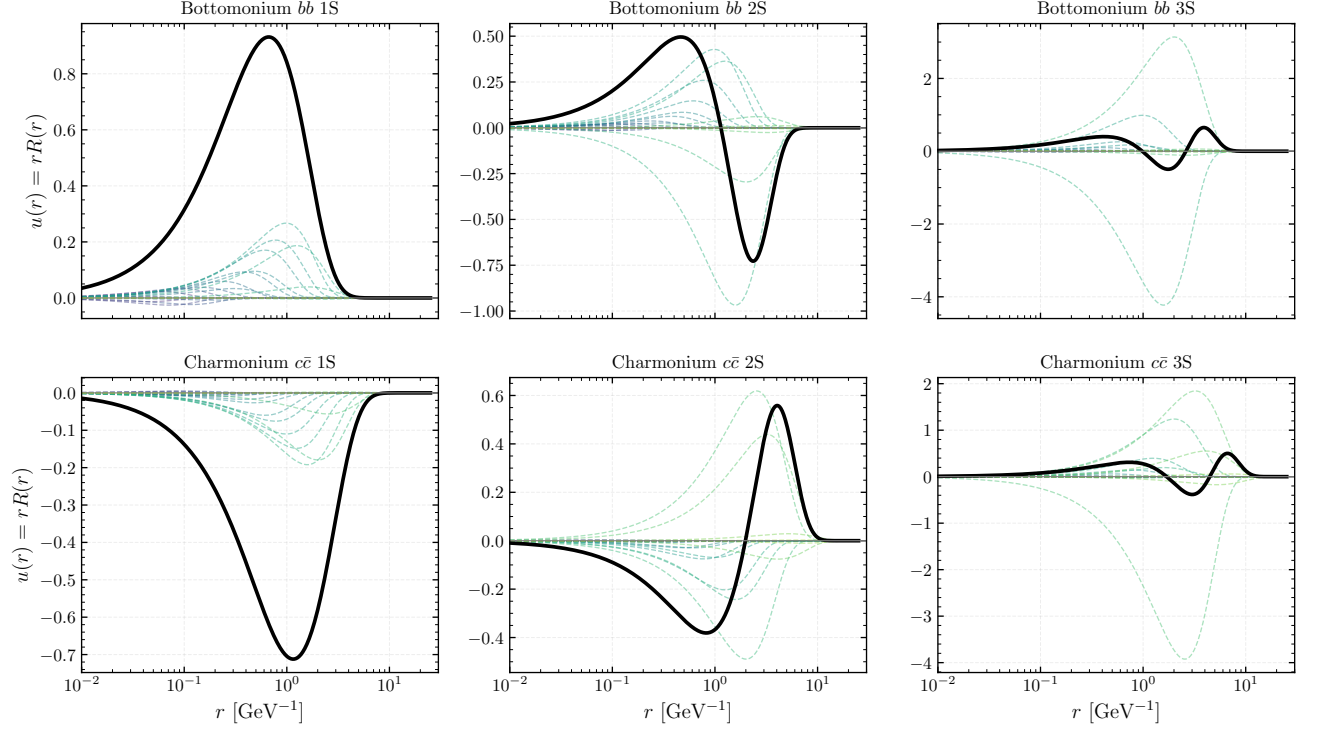


FIG. 2. Gaussian-basis coverage of the 1S, 2S, and 3S radial amplitudes for bottomonium (top row) and charmonium (bottom row). Dashed curves are the weighted basis functions, the solid curve their sum. The 3S panels make the partial linear dependence of the non-orthogonal basis visible as large, alternating coefficients (Appendix D).

Reconstructed S-wave radial amplitudes (radial nodes visible)

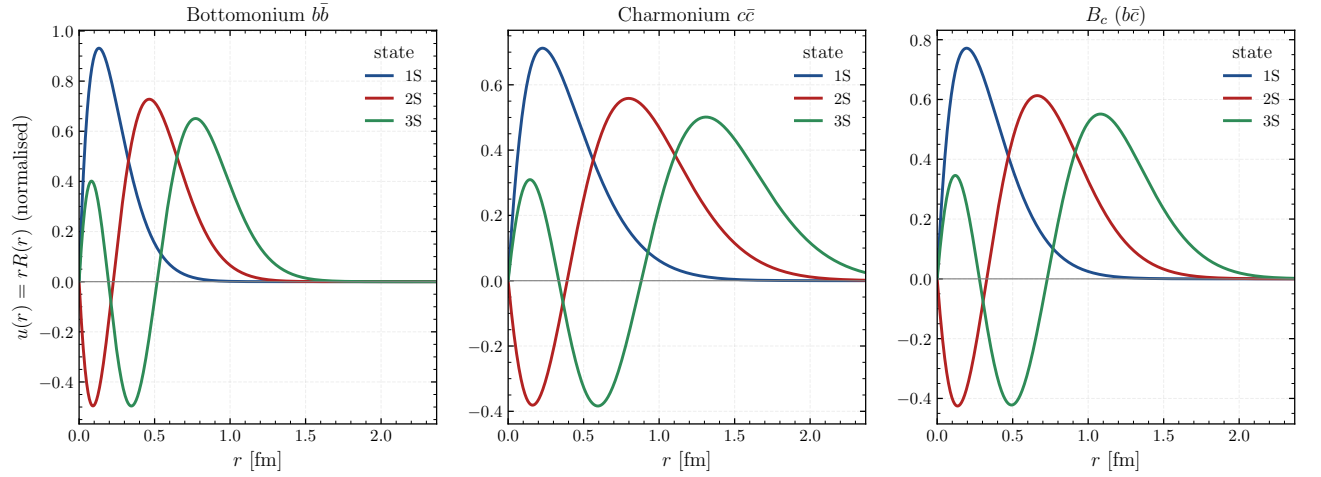


FIG. 3. Reconstructed S -wave radial amplitudes $u(r) = rR(r)$ for the 1S, 2S, and 3S states of the $b\bar{b}$, $c\bar{c}$, and $b\bar{c}$ sectors, showing the expected 0, 1, and 2 radial nodes. The heavier $b\bar{b}$ system is the most tightly bound (smallest spatial extent), consistent with its smaller RMS radius (Table VII).

treating the parameters as the dominant, approximately independent sources (the off-diagonal correlations of α_s and b are sub-leading for the masses and are neglected in the quoted error). This answers “how far does the prediction move when the fitted parameters wiggle within

their errors,” and it is what appears as the error bars in the figures and the $\pm$ in the tables.

The second is the *physical* error bar σ_i that enters the χ^2 and the pulls. It is *not* the model covariance. The PDG mass errors of the lowest states are 0.1–1 MeV,

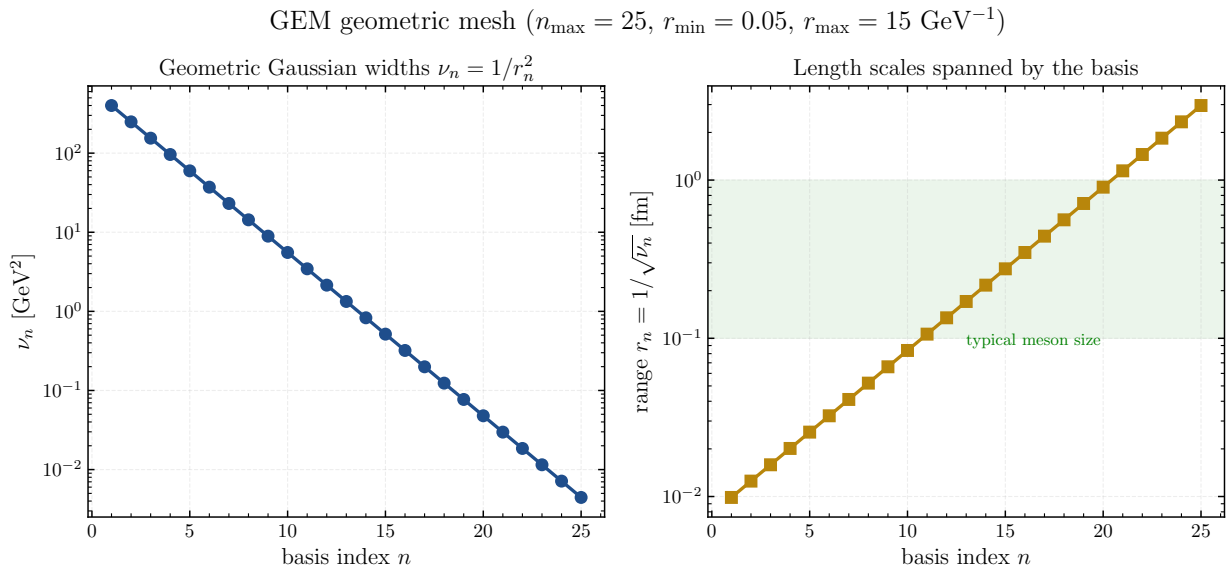


FIG. 4. The geometric Gaussian mesh of Eq. (12). Left: the basis widths $\nu_n = 1/r_n^2$ in geometric progression. Right: the corresponding physical ranges $r_n = 1/\sqrt{\nu_n}$ in fm, spanning from well inside the one-gluon-exchange cusp to far into the confinement tail and bracketing the typical meson size (shaded). The mesh is fixed in advance ($n_{\max} = 25$, $r_{\min} = 0.05$, $r_{\max} = 15 \text{ GeV}^{-1}$); only the linear coefficients are varied.

far below the intrinsic accuracy of a quenched Cornell-plus-perturbative model. The honest uncertainty on a predicted mass is dominated by a *theory systematic*—the neglected $\mathcal{O}(v^2/c^2)$ relativistic physics, coupled channels, and quenching—which we estimate as $\sigma_{\text{theory}} = 10 \text{ MeV}$ and combine in quadrature with the experimental error,

$$\sigma_i = \sqrt{(\sigma_i^{\text{exp}})^2 + \sigma_{\text{theory}}^2}. \quad (21)$$

On the annihilation widths the leading-order non-relativistic formulae carry large perturbative-QCD and relativistic corrections, so we assign a 30% fractional theory error. Using Eq. (21) in the objective makes the fit essentially a uniform-weight least squares in which the precisely measured J/ψ and Υ no longer swamp the rest, without any *ad hoc* per-state weighting.

B. Reduced χ^2 and pulls

With Eq. (21), a reduced $\chi^2/\text{dof} \approx 1$ means the model reproduces the spectrum to within its intrinsic systematic, and $\chi^2/\text{dof} \gg 1$ flags a sector where the neglected physics genuinely exceeds the floor. The degrees of freedom are $\text{dof} = N - N_{\text{free}}$, counting only the genuinely varied parameters: a tightly bounded (frozen) parameter, such as the interpolated B_c coupling or the fixed D -sector inputs, is not a degree of freedom. For the B_c ($N = 2$, two free) and D ($N = 1$, one free) sectors $\text{dof} = 0$: these are exactly determined, so a reduced χ^2 is undefined and is reported as “—”, never as a negative number. The model-independent figure of merit, which assumes nothing about σ_i , is the RMS mass deviation

$\text{RMS} = [\frac{1}{N} \sum_i (M_i^{\text{calc}} - M_i^{\text{exp}})^2]^{1/2}$, which we quote first. The dimensionless pull of each state, $(M_i^{\text{calc}} - M_i^{\text{exp}})/\sigma_i$, localizes the tension (Fig. 12).

C. Cross-validation against overfitting

A four-parameter fit to ~ 12 masses could in principle be *memorizing* the training levels rather than learning the dynamics. We test this directly by leave-one-out (LOO) cross-validation: for each measured level we refit the potential on every *other* level and predict the held-out one; the RMS of those held-out predictions (RMS_{LOO}) is the honest out-of-sample error. Comparing it with the in-sample full-fit RMS, $\text{RMS}_{\text{LOO}} \approx \text{RMS}_{\text{full}}$ signals a model that generalizes, whereas $\text{RMS}_{\text{LOO}} \gg \text{RMS}_{\text{full}}$ would be the fingerprint of memorization. The results (Table I) give a LOO-to-full ratio of ~ 2 in both quarkonium sectors—a mild and expected inflation with so few levels per parameter, and far from the runaway that overfitting would produce. The rigid, physically constrained Cornell form therefore generalizes; the larger charmonium deviations (Sec. VIA) are physics, not overfitting. We use LOO rather than a 2-fold split because each refit retains $N - 1$ well-anchored levels; a half-split can accidentally drop the ground-state anchors and explode for reasons unrelated to overfitting. The many refits are cheap because the GEM eigenvalues are built from analytic integrals over the Gaussian widths and are independent of the radial grid used only for output wavefunctions.

TABLE I. Leave-one-out cross-validation of the mass fit. The out-of-sample (LOO) RMS is close to the in-sample full-fit RMS, with a ratio near 2 that is expected for so few levels per free parameter; this rules out overfitting.

Sector	N levels	RMS _{full} (MeV)	RMS _{LOO} (MeV)	LOO/full
Bottomonium ($b\bar{b}$)	12	5.57	11.06	1.99
Charmonium ($c\bar{c}$)	10	14.14	30.99	2.19

IV. DECAY WIDTHS

The wavefunctions obtained above are used to evaluate a set of decay observables. Except where noted these are leading-order non-relativistic expressions; their accuracy is discussed in Sec. VID.

A. Leptonic Decay Widths

For a vector meson the process $V \rightarrow e^+e^-$ proceeds through quark–antiquark annihilation into a virtual photon. The van Royen–Weisskopf formula [17], with the leading-order QCD radiative correction, gives

$$\Gamma(V \rightarrow e^+e^-) = \frac{4\alpha_{\text{em}}^2 e_q^2}{M^2} |R(0)|^2 \left(1 - \frac{16\alpha_s}{3\pi}\right), \quad (22)$$

where e_q is the quark electric charge in units of e ($e_b = -1/3$, $e_c = +2/3$), M is the meson mass, and $R(0)$ is the radial wavefunction at the origin. The coupling in the radiative factor is evaluated at the annihilation scale $Q \sim M$, which differs from the effective long-distance α_s of the Cornell potential; we use $\alpha_s \simeq 0.20$ for bottomonium and $\simeq 0.35$ for charmonium, consistent with the running coupling at those scales. With $\alpha_s \simeq 0.35$ the radiative factor is itself a $\sim 60\%$ correction, signalling that the perturbative series for this observable is only marginally convergent in the charmonium sector. The 3D_1 states relevant to e^+e^- couple through the second derivative $R''(0)$ of the wavefunction, evaluated directly from the basis, and the physical $2S$ – $1D$ states are treated as the S – D mixtures of Sec. IA 4.

B. Two-Photon Decay Width

For a pseudoscalar ($S = 0$) state the analogous annihilation into two photons gives

$$\Gamma(P \rightarrow \gamma\gamma) = \frac{12\alpha_{\text{em}}^2 e_q^4}{M^2} |R(0)|^2 \left(1 - \frac{20 - \pi^2}{3} \frac{\alpha_s}{\pi}\right). \quad (23)$$

The charge enters as e_q^4 because the process involves two electromagnetic vertices. The coefficient $(20 - \pi^2)/3 \simeq 3.38$ is the next-to-leading-order perturbative-QCD correction for the $^1S_0 \rightarrow \gamma\gamma$ channel [2]. For the χ_0 and

χ_2 ($^3P_{0,2}$) states the two-photon width is governed instead by the derivative of the wavefunction at the origin, $|R'(0)|^2$, through the Barbieri–Gatto–Remiddi formula [18]; the χ_1 (3P_1) channel is forbidden by the Landau–Yang theorem.

C. Radiative Transitions: $M1$ and $E1$

Magnetic-dipole ($M1$) transitions connect spin-triplet vector mesons to spin-singlet pseudoscalars, e.g. $J/\psi \rightarrow \eta_c\gamma$, through the quark magnetic moment [19]:

$$\Gamma(V \rightarrow P + \gamma) = \frac{4\alpha_{\text{em}} e_q^2}{3} k^3 \left(\frac{1}{2m_1} + \frac{1}{2m_2}\right)^2 |\langle f|i \rangle|^2, \quad (24)$$

where $k = (M_i^2 - M_f^2)/2M_i$ is the photon energy and $\langle f|i \rangle = \sum_{jk} C_j^f C_k^i \mathcal{I}_2(\nu_j^f + \nu_k^i)$ is the spatial overlap of the initial and final states. Because the hyperfine interaction makes the vector and pseudoscalar radial wavefunctions slightly different, this overlap is not exactly unity. Electric-dipole ($E1$) transitions connect P - and S -wave states [19], e.g. $\chi_{cJ} \rightarrow J/\psi\gamma$:

$$\Gamma(P \rightarrow S + \gamma) = \frac{4\alpha_{\text{em}} e_q^2}{9} k^3 |\langle f|r|i \rangle|^2, \quad (25)$$

with the dipole matrix element $\langle f|r|i \rangle = \sum_{jk} C_j^f C_k^i \mathcal{I}_4(\nu_j^f + \nu_k^i)$. The moment is $\mathcal{I}_4$, not $\mathcal{I}_2$, because the explicit factor of r from the dipole operator raises the power of the integrand by one relative to a plain overlap—a point we emphasize because dropping it underestimates the $E1$ rates by almost an order of magnitude. Equations (24) and (25) are the spin-averaged leading-order rates; they omit relativistic recoil of the final-state meson and the fine-structure-dependent statistical factors, so the J -dependence of the calculated $\chi_J \rightarrow V\gamma$ widths enters only through the photon energy k^3 . Parameter uncertainties are propagated to all widths by finite differences, Eq. (20).

D. Open-Flavour Decays: the 3P_0 Model

Open-flavour strong decays, such as $\psi(3770) \rightarrow D\bar{D}$, are estimated with the 3P_0 quark-pair-creation model [20, 21], in which a $q\bar{q}$ pair is created from the vacuum with 3P_0 quantum numbers and recombines with the initial constituents into two final mesons. The width has the schematic form

$$\Gamma = 2\pi p \frac{\sqrt{M_B^2 + p^2} \sqrt{M_C^2 + p^2}}{M_A} \gamma^2 |\mathcal{O}|^2, \quad (26)$$

where p is the final-state momentum, γ is the dimensionless pair-creation strength, and $\mathcal{O}$ is a spatial overlap of the initial and final GEM wavefunctions. A fully consistent evaluation of $\mathcal{O}$ requires a momentum-space overlap

in generalized Jacobi coordinates including the P -wave creation vertex and the spin-flavour-colour recoupling; the present implementation uses a simplified coordinate-space overlap and fixes γ to the measured $\psi(3770)$ total width. We therefore treat the 3P_0 sector as a showcase rather than a prediction (Sec. VID).

E. Wavefunction at the Origin

The annihilation widths of Eqs. (22) and (23) require the radial wavefunction at the origin, which is nonzero only for S -waves. Rather than summing the Gaussian coefficients directly—numerically delicate because individual basis functions are large at $r = 0$ —we use the hypervirial (Schwinger) relation, which for an S -wave eigenstate gives

$$|R(0)|^2 = \frac{2\mu}{\hbar^2} \left\langle \frac{dV}{dr} \right\rangle = 2\mu \left[\frac{4\alpha_s}{3} \langle r^{-2} \rangle + b \right], \quad (27)$$

where $\langle r^{-2} \rangle$ is evaluated analytically from the Gaussian basis. The relation returns the origin density as a stable expectation value, which matters because every S -wave annihilation width leans on it. For the 3D_1 states the analogous quantity is $|R''(0)|^2$, obtained directly from the basis.

V. COMPUTATIONAL IMPLEMENTATION

A. Validation: the Quartic Anharmonic Oscillator

Before applying the solver to the meson sector we validated the variational machinery on a problem with a numerically exact answer: the one-dimensional quartic anharmonic oscillator, $V(x) = \frac{1}{2}m\omega^2x^2 + \lambda x^4$, in the strongly anharmonic regime $\lambda = 100$ (in units $m = \omega = \hbar = 1$), where naive perturbation theory diverges. Table II compares the lowest two eigenvalues from (i) Rayleigh–Schrödinger perturbation theory, (ii) a single-Gaussian variational ansatz, (iii) a flexible Gaussian-times-polynomial variational ansatz of the same family as the GEM, (iv) a finite-difference matrix diagonalization, and (v) a shooting integration of the Schrödinger ODE. The flexible variational ansatz reproduces the finite-difference and shooting eigenvalues to better than 10^{-4} , while the single Gaussian gives a strict upper bound $\sim 2\%$ above and perturbation theory fails outright. This confirms the variational principle, the matrix-element machinery, and the eigenvalue solver on which the meson calculation rests.

B. Virial-Theorem Consistency Check

As an internal check on every converged meson eigenstate, we verify the quantum virial theorem, $2\langle T \rangle =$

TABLE II. Lowest two eigenvalues of the quartic anharmonic oscillator ($\lambda = 100$, $m = \omega = \hbar = 1$) from independent methods. The flexible Gaussian-polynomial variational ansatz agrees with the numerically exact finite-difference and shooting solutions to $\sim 10^{-5}$; second-order perturbation theory diverges in this strongly coupled regime.

Method	E_0	E_1
Perturbation theory (2nd order)	−26174.5	−205873.5
Variational (single Gaussian)	3.19243	11.32478
Variational (Gaussian×poly)	3.13138	11.18745
Finite-difference matrix	3.13138	11.18723
Shooting (ODE)	3.13138	11.18725

$\langle r dV/dr \rangle$. For the Cornell potential $r dV/dr = (4\alpha_s/3)r^{-1} + br$ (the constant c drops out), and including the smeared hyperfine contribution $r dV_{SS}/dr = -2\sigma^2 r^2 V_{SS}$ the ratio

$$\mathcal{R} = \frac{2\langle T \rangle}{\left\langle \frac{4\alpha_s}{3r} + br + r \frac{dV_{SS}}{dr} \right\rangle} \quad (28)$$

must equal unity for a complete, well-converged variational basis. For the fitted potentials of all sectors $\mathcal{R} = 0.99999$ – 1.00000 for the S -wave ground states, confirming the convergence of the $n_{\max} = 25$ basis; every downstream observable is computed from eigenstates that pass this test.

C. Solver Structure and Pipeline

The complete computational pipeline is summarized in Fig. 5. All reference data are first read live from the PDG SQLite database (masses, e^+e^- widths, and experimental errors). For each sector and each (L, S) channel the geometric Gaussian basis is laid down and the Hamiltonian and overlap matrices are assembled in closed form from the Gamma-function moments of Sec. II A; the generalized eigenvalue problem is then solved, the perturbative spin-orbit and tensor shifts and the 3S_1 – 3D_1 mixing are applied, and the model masses and leptonic widths are formed. These feed the physical- σ χ^2 objective of Eq. (18), which a trust-region least-squares optimizer minimizes by updating (α_s, b, c, σ) and reassembling the matrices until convergence (the B_c coupling and smearing enter frozen, from the renormalization-group interpolation of Eq. (19)). Once the parameters are fixed, the converged wavefunctions yield the static observables and decay widths of Sec. IV; the fitted-parameter covariance is propagated to every one of them by finite differences [Eq. (20)]; the fit is stress-tested for overfitting by leave-one-out cross-validation (Sec. III); and the spectra, widths, report, and figures are exported.

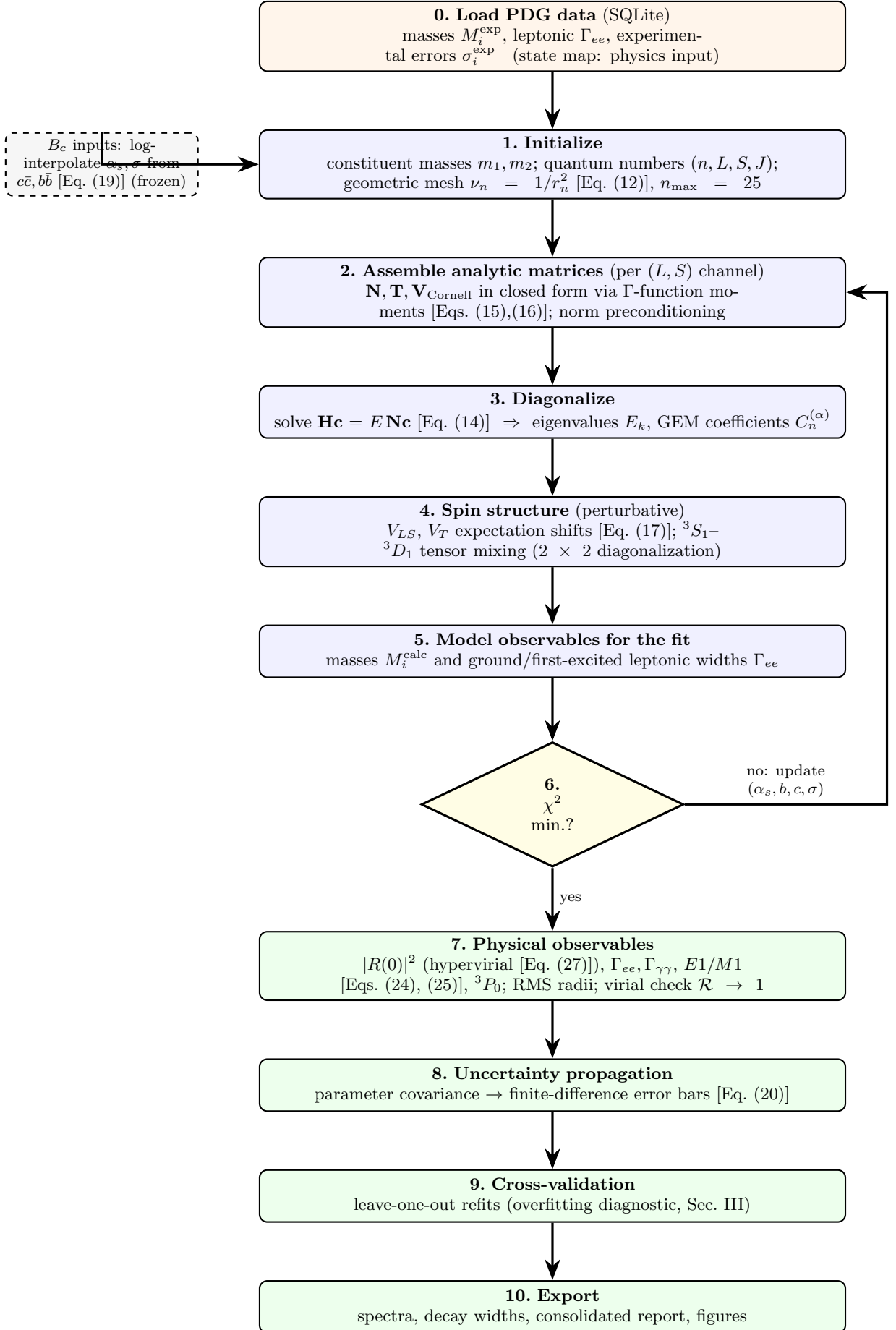


FIG. 5. Computational pipeline of the heavy-meson GEM solver, from the live PDG data load to the exported observables. Closed-form matrix assembly (step 2) and diagonalization (step 3) sit inside the trust-region χ^2 loop (step 6), which reassembles the analytic matrices for each trial (α_s, b, c, σ) until the physical- σ objective of Eq. (18) is minimized; the B_c coupling and smearing enter frozen from the renormalization-group interpolation [Eq. (10)]. Once the parameters converge, the wavefunctions

VI. RESULTS AND DISCUSSION

A. Mass Spectra

The fitted spectra are compared with the PDG values in Tables IV (charmonium), V (bottomonium), and VI (B_c and D); the global goodness of fit is summarized in Table III. All states listed with an experimental value were included as targets in the χ^2 fit, so the agreement measures the flexibility of the Cornell-plus-Breit-Fermi form rather than out-of-sample predictive power—which is instead quantified by the cross-validation of Sec. III.

TABLE III. Global goodness of fit per sector with the physical error model of Eq. (21). “RMS” is the model-independent RMS mass deviation. The B_c and D sectors are exactly determined (dof = 0), so a reduced χ^2 is undefined.

Sector	N	χ^2	dof	χ^2/dof	RMS (MeV)
Bottomonium ($b\bar{b}$)	14	6.70	10	0.67	7.0
Charmonium ($c\bar{c}$)	14	102.72	10	10.27	28.8
B_c ($b\bar{c}$)	2	0.00	0	—	0.0
D ($c\bar{u}$)	1	0.00	0	—	0.0
Global	31	109.4	20	5.47	—

We reproduce the bottomonium spectrum very well: the RMS deviation is only 7.0 MeV and $\chi^2/\text{dof} = 0.67$, so the model describes the $b\bar{b}$ levels to within the assumed 10 MeV theory systematic. No single $b\bar{b}$ level is off by more than 14 MeV, and the worst pull, -1.4σ , belongs to the 1^3D_2 state $\Upsilon_2(1D)$. The charmonium spectrum comes out less accurately—RMS 28.8 MeV, $\chi^2/\text{dof} = 10.3$ —and here the deviation is strongly state-dependent: the S - and P -wave levels still lie within ~ 20 MeV of experiment, but the high D -wave and the radially excited states are what drive the χ^2 . The single largest tension is the 2^3D_1 assignment to $\psi(4160)$, which we predict 79 MeV too low (-7.0σ), followed by the 1^3D_3 $\psi_3(3842)$ (-48 MeV) and the 1^3D_2 $\psi_2(3823)$ (-30 MeV). This pattern is exactly the expected signature of the larger charm velocity ($v^2 \approx 0.3$): the high- L , high- n states sample the long-range, relativistically corrected part of the interaction most strongly, and that is just where the quenched static Cornell form is least adequate. It is physics, not a fitting artefact—the cross-validation confirms that our parameters generalize (Sec. III). The fitted couplings $\alpha_s^{c\bar{c}} = 0.536 > \alpha_s^{b\bar{b}} = 0.349$ (Appendix B) follow the expected running-coupling trend, with a larger effective α_s at the lower charm scale. We show the full level diagrams, with the model uncertainty as error bars against the PDG levels, in Fig. 6.

For the B_c sector the two measured $L = 0$ masses fix b and c exactly, so the small ground-state deviations are not predictions; the genuine prediction is the rest of the multiplet (Sec. VIB and Appendix C). The predicted B_c^* (1^3S_1) mass, 6.346 GeV, lies 16 MeV above the HPQCD lattice value 6.330(7) GeV [22], a $\sim 0.25\%$ difference.

The heavy-light D sector is anchored by the single D^0 mass; the predicted D^{*0} vector, 1.984 GeV, undershoots the PDG 2.007 GeV by 23 MeV (1.1%), the largest fractional miss of any ground state and a direct consequence of the non-relativistic treatment being least controlled when one constituent is light.

B. Comparison with the Literature

Figure 7 places our masses against six contemporary quark-model calculations for charmonium and five for bottomonium: the nonrelativistic Cornell model of Soni *et al.* [9], the relativistic quasipotential model of Ebert, Faustov, and Galkin [23], the screened relativized model of Deng *et al.* [24], the one-loop Cornell model of Radford and Repko [25], the nonrelativistic and relativized Godfrey-Isgur models of Barnes, Godfrey, and Swanson [8], and (for bottomonium) Godfrey and Moats [26]. Plotted as the deviation $M_{\text{model}} - M_{\text{exp}}$, our results cluster within the ± 10 MeV theory band for the low-lying states and spread out for the $c\bar{c}$ D -waves, just as the discussion above leads us to expect. Figure 8 then aggregates this into an RMS accuracy per model over a common set of experimentally known states; we hatch our own work there because it is *fitted* to those masses, whereas the literature values are genuine predictions, so the meaningful comparison is the one among the predictive models.

The B_c spectrum is the most predictive output, since only the two $L = 0$ ground states are measured. Figure 9 compares the full predicted B_c multiplet with the relativistic [23], relativized Godfrey-Isgur, and Cornell quark-model calculations collected in Ref. [9], and with the HPQCD lattice point for the B_c^* [22]. The present B_c levels track the model average closely; the full table is given in Appendix C. The corresponding D -sector comparison with the relativistic model of Ref. [27] is shown in Fig. 10.

C. Fit Quality and Localized Discrepancies

Figures 11 and 12 show the residuals and the pulls per state. In bottomonium every pull lies within $\sim 1.4\sigma$; in charmonium the S - and P -waves are comparable, while the D -wave multiplet and the $2^3P_2/\psi(4160)$ assignments carry the few large pulls already identified. We stress that these are not failures of the solver: the same potential reproduces the partner states in the same multiplet at the few-MeV level, and the discrepancies grow monotonically with L and n , exactly as the non-relativistic expansion predicts. The reduced χ^2 and the cross-validation are juxtaposed in Fig. 13, which visualizes the central statistical message: bottomonium sits at $\chi^2/\text{dof} \approx 0.7$, charmonium well above unity, and in both cases the held-out (LOO) RMS stays close to the in-sample RMS.

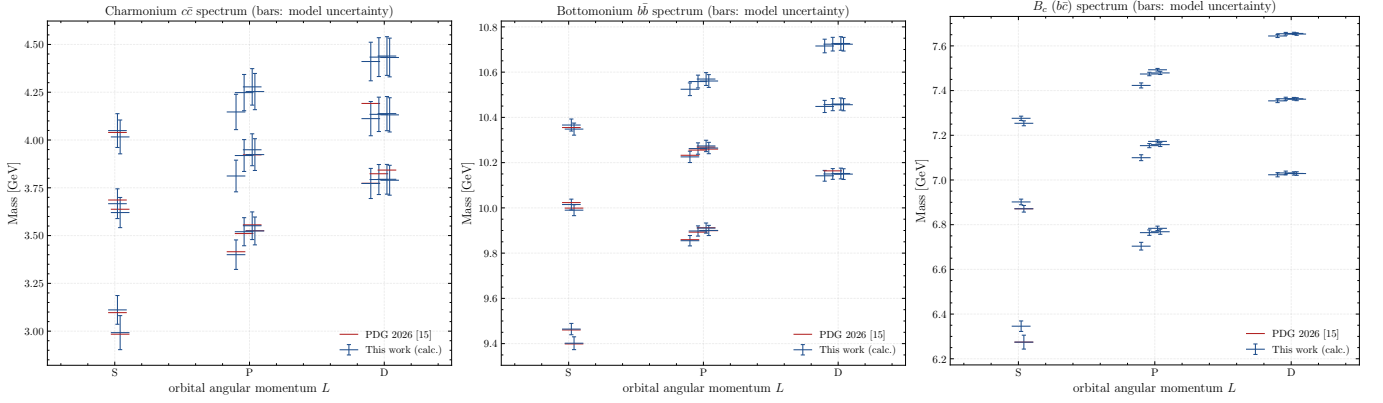


FIG. 6. Grotrian-style level diagrams for charmonium (left), bottomonium (centre), and B_c (right): calculated masses with the propagated model uncertainty (blue) against the PDG values (red), grouped by orbital angular momentum L . For B_c only the two $L = 0$ ground states are measured; the remaining levels are predictions.

TABLE IV. Calculated charmonium ($c\bar{c}$) masses compared with PDG data. All listed states were included in the χ^2 fit. “Pull” is $(M^{\text{calc}} - M^{\text{exp}})/\sigma$ with the physical σ of Eq. (21). The quoted $\pm$ is the propagated model uncertainty.

State	J^{PC}	Calc. (GeV)	Exp. (GeV)	Δ (MeV)	Pull
1^1S_0 (η_c)	0^{-+}	2.992(89)	2.9841	+8.2	+0.82
1^3S_1 (J/ψ)	1^{--}	3.111(75)	3.0969	+14.3	+1.43
1^3P_0 (χ_{c0})	0^{++}	3.400(77)	3.4155	-15.4	-1.54
1^3P_1 (χ_{c1})	1^{++}	3.520(73)	3.5107	+9.6	+0.96
1^1P_1 (h_c)	1^{+-}	3.524(73)	3.5254	-1.1	-0.11
1^3P_2 (χ_{c2})	2^{++}	3.552(72)	3.5562	-4.6	-0.46
2^1S_0 ($\eta_c(2S)$)	0^{-+}	3.621(79)	3.6378	-17.2	-1.72
2^3S_1 ($\psi(2S)$)	1^{--}	3.667(78)	3.6861	-19.3	-1.93
1^3D_1 ($\psi(3770)$)	1^{--}	3.773(79)	3.7737	-1.1	-0.10
1^3D_2 ($\psi_2(3823)$)	2^{--}	3.793(79)	3.8235	-30.2	-3.02
1^3D_3 ($\psi_3(3842)$)	3^{--}	3.795(78)	3.8427	-48.2	-4.82
2^3P_2 ($\chi_{c2}(3930)$)	2^{++}	3.949(83)	3.9225	+26.6	+2.65
3^3S_1 ($\psi(4040)$)	1^{--}	4.049(88)	4.0396	+9.7	+0.90
2^3D_1 ($\psi(4160)$)	1^{--}	4.112(90)	4.1913	-79.4	-7.05

D. Decay Widths and Static Observables

Static observables (RMS radii and the wavefunction at the origin) are collected in Table VII and Figs. 16 and 17; the leptonic and two-photon annihilation widths in Table VIII; and the radiative $E1/M1$ transitions in Table IX. The corresponding comparison figures are Figs. 14–18.

Our leptonic widths agree with experiment at the 11–14% level for the ground- and first-excited vectors of both families (J/ψ : 4.90 vs. 5.53 keV; $\Upsilon(1S)$: 1.51 vs. 1.34 keV), and the pulls stay below 0.5σ once we apply the 30% theory error (Fig. 14). The agreement degrades for the $3S$ vector $\psi(4040)$ (+66%) and breaks down completely for the D -wave vectors $\psi(3770)$ and $\psi(4160)$, whose tiny e^+e^- widths (~ 0.08 and ~ 0.02 keV) are governed by $R''(0)$ and by the S – D mixing angle, both of which are poorly determined at this order. The χ_{c0} and χ_{c2} two-photon widths come out well (+11% for χ_{c0} , +4% for χ_{c2} , both through $|R'(0)|^2$), but we overpredict the S -wave pseudoscalar η_c by a factor of 2.4 (15.4 vs. 6.4 keV) and $\eta_c(2S)$ by 2.3.

The contrast between the leptonic and two-photon S -wave results is informative and resolves an apparent paradox. Both widths scale with the S -wave origin density, yet the η_c two-photon width is 2.4 times too large while the J/ψ leptonic width is only 11% too small. The resolution is the hyperfine interaction. The spin-singlet η_c feels an attractive contact term ($\langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle = -3/4$) that pulls its wavefunction inward, raising $|R(0)|^2$ to 2.09 GeV^3 —roughly twice the Eichten–Quigg reference [30], Table VII—whereas the spin-triplet J/ψ (+1/4) is less concentrated, with a hypervirial $|R(0)|^2 \approx 1.2 \text{ GeV}^3$ close to that reference. The two-photon overestimate is therefore not a generic “ $|R(0)|^2$ is too big” but specifically the non-relativistic over-concentration of the spin-singlet at the origin, compounded by the leading-order treatment of a short-distance observable. Figure 16 makes this excess visible by overlaying our $|R(0)|^2$ on the Eichten–Quigg values. A quantitative description of these annihilation rates requires the $\mathcal{O}(v^2/c^2)$ corrections of the NRQCD factorization framework [1, 2], including relative-momentum operators at the annihilation vertex, which are beyond the present leading order.

TABLE V. Calculated bottomonium ($b\bar{b}$) masses compared with PDG data. All listed states were included in the χ^2 fit; the larger quark mass makes the non-relativistic treatment more accurate.

State	J^{PC}	Calc. (GeV)	Exp. (GeV)	Δ (MeV)	Pull
1^1S_0 (η_b)	0^{-+}	9.402(28)	9.3987	+3.1	+0.30
1^3S_1 ($\Upsilon(1S)$)	1^{--}	9.464(25)	9.4604	+3.8	+0.38
1^3P_0 ($\chi_{b0}(1P)$)	0^{++}	9.855(23)	9.8594	-4.6	-0.46
1^3P_1 ($\chi_{b1}(1P)$)	1^{++}	9.898(23)	9.8928	+5.1	+0.51
1^1P_1 ($h_b(1P)$)	1^{+-}	9.900(22)	9.8993	+1.0	+0.09
1^3P_2 ($\chi_{b2}(1P)$)	2^{++}	9.911(22)	9.9122	-1.5	-0.15
2^1S_0 ($\eta_b(2S)$)	0^{-+}	9.990(24)	9.9990	-9.1	-0.83
2^3S_1 ($\Upsilon(2S)$)	1^{--}	10.015(24)	10.0234	-8.6	-0.86
1^3D_2 ($\Upsilon_2(1D)$)	2^{--}	10.150(24)	10.1637	-13.8	-1.37
2^3P_0 ($\chi_{b0}(2P)$)	0^{++}	10.225(25)	10.2325	-7.0	-0.70
2^3P_1 ($\chi_{b1}(2P)$)	1^{++}	10.262(25)	10.2555	+6.8	+0.68
2^1P_1 ($h_b(2P)$)	1^{+-}	10.265(25)	10.2598	+4.8	+0.47
2^3P_2 ($\chi_{b2}(2P)$)	2^{++}	10.274(25)	10.2686	+5.1	+0.51
3^3S_1 ($\Upsilon(3S)$)	1^{--}	10.366(27)	10.3552	+11.1	+1.11

TABLE VI. Mixed-flavour B_c ($b\bar{c}$) and heavy-light D ($c\bar{u}$) sectors. States with an experimental mass were used as the fit anchors (the B_c b, c parameters and the D offset c); the remaining entries are predictions. The B_c^* row is compared with the HPQCD lattice value [22], the D^{*0} row with the PDG mass (not used in the fit).

Sector	State	Calc. (GeV)	Ref. (GeV)
$b\bar{c}$	1^1S_0 ($B_c^\pm$)	6.274(31)	6.2745 (fit)
$b\bar{c}$	1^3S_1 ($B_c^{*\pm}$)	6.346(23)	6.330(7) (latt.)
$b\bar{c}$	2^1S_0 ($B_c(2S)^\pm$)	6.871(15)	6.8712 (fit)
$b\bar{c}$	1^1P_1 (pred.)	6.768(11)	—
$b\bar{c}$	1^3D_1 (pred.)	7.023(9)	—
$c\bar{u}$	1^1S_0 (D^0)	1.864(0)	1.8648 (fit)
$c\bar{u}$	1^3S_1 (D^{*0})	1.984(0)	2.0069 (PDG)

TABLE VII. Static observables from the GEM wavefunctions: RMS radius and the spin-singlet S -wave density at the origin $|R(0)|^2$ (which drives the two-photon widths). The Eichten–Quigg reference [30] is shown for the ground states.

Sector	State	$\sqrt{\langle r^2 \rangle}$ (fm)	$ R(0) ^2$ (GeV ³)	EQ [30]
$c\bar{c}$	$1S$	0.363	2.088	1.095
$c\bar{c}$	$2S$	0.840	0.882	0.697
$c\bar{c}$	$1P$	0.694	—	—
$b\bar{b}$	$1S$	0.210	12.15	5.859
$b\bar{b}$	$2S$	0.492	5.04	2.897
$b\bar{b}$	$1P$	0.402	—	—
$b\bar{c}$	$1S$	0.306	3.62	—

The radiative transitions are the most successful decay observables (Table IX, Fig. 18). The $E1$ widths $\chi_{cJ}(1P) \rightarrow J/\psi \gamma$ reproduce the measured pattern and ordering: χ_{c0} (134 vs. ~ 116 keV), χ_{c1} (361 vs. ~ 288 keV), χ_{c2} (445 vs. ~ 406 keV), all within 15–25%, and the bottomonium $\chi_{bJ}(1P) \rightarrow \Upsilon \gamma$ widths agree at the 10% level (χ_{b0} : 25.3 vs. 27.5 keV; χ_{b1} : 34.4 vs. 31.1 keV). This level of agreement depends on the correct dipole matrix element $\langle f|r|i \rangle$ of Eq. (25): using a plain overlap instead of the $\mathcal{I}_4$ moment underestimates these $E1$ widths

TABLE VIII. Leptonic (e^+e^-) and two-photon ($\gamma\gamma$) annihilation widths compared with PDG data. The η_c two-photon width is overpredicted (see text); the S -wave leptonic widths agree at the 10–15% level.

Channel	State	Calc. (keV)	Exp. (keV)	Pull
e^+e^-	J/ψ	4.90	5.53	+0.34
e^+e^-	$\psi(2S)$	1.98	2.29	+0.44
e^+e^-	$\psi(4040)$	1.38	0.83	+1.96
e^+e^-	$\Upsilon(1S)$	1.51	1.34	+0.40
e^+e^-	$\Upsilon(2S)$	0.65	0.61	+0.23
e^+e^-	$\Upsilon(3S)$	0.48	0.44	+0.30
$\gamma\gamma$	η_c	15.4	6.4	+2.24
$\gamma\gamma$	$\eta_c(2S)$	4.7	2.0	+3.24
$\gamma\gamma$	χ_{c0}	2.18	2.43	+0.32
$\gamma\gamma$	χ_{c2}	0.58	0.56	+0.11
$\gamma\gamma$	η_b	0.77	—	—

by almost an order of magnitude. The $M1$ widths are smaller and more uncertain— $J/\psi \rightarrow \eta_c \gamma$ comes out at 2.98 keV against the measured 1.58 keV—as expected, since $M1$ rates depend on the small vector–pseudoscalar overlap defect and on relativistic corrections that are numerically important for these hindered transitions.

TABLE IX. Calculated $E1$ and $M1$ radiative transition widths with propagated model uncertainty, compared with experiment. The χ_{b2} transition has no clean absolute experimental width.

Type	Transition	Calc. (keV)	Exp. (keV)
$M1$	$J/\psi \rightarrow \eta_c \gamma$	2.98(2.37)	1.58
$M1$	$\psi(2S) \rightarrow \eta_c(2S) \gamma$	0.18(16)	~ 0.05
$E1$	$\chi_{c0} \rightarrow J/\psi \gamma$	133.7(9.7)	~ 116
$E1$	$\chi_{c1} \rightarrow J/\psi \gamma$	361.4(35.0)	~ 288
$E1$	$\chi_{c2} \rightarrow J/\psi \gamma$	445.1(53.5)	~ 406
$M1$	$\Upsilon(1S) \rightarrow \eta_b \gamma$	0.012(5)	~ 0.01
$E1$	$\chi_{b0} \rightarrow \Upsilon(1S) \gamma$	25.3(0.8)	~ 27.5
$E1$	$\chi_{b1} \rightarrow \Upsilon(1S) \gamma$	34.4(1.3)	~ 31.1
$E1$	$\chi_{b2} \rightarrow \Upsilon(1S) \gamma$	37.5(1.5)	—

Finally, the 3P_0 open-flavour showcase: tuning the sin-

gle pair-creation strength to the measured $\psi(3770)$ total width gives $\gamma = 1.16$ and, by construction, reproduces $\Gamma(\psi(3770) \rightarrow D\bar{D}) = 27.2$ MeV. Carrying that same γ to the 2^3D_1 state $\psi(4160)$ predicts $\Gamma(\psi(4160) \rightarrow D\bar{D}) = 67.8 \pm 11.9$ MeV. Because the overlap is the simplified coordinate-space form of Sec. IV D and γ is fitted rather than predicted, we present this as an illustration of the machinery (Fig. 19), not as a quantitative prediction.

E. Physics Inputs: the Fitted Potentials

The four fitted Cornell potentials are shown in Fig. 20, and Fig. 22 compares their shape and string tension with the canonical Eichten form [4] and the accepted lattice/Regge value $\sigma \approx 0.18$ GeV². The fitted string tensions ($b = 0.195$ GeV² for $b\bar{b}$, 0.133 GeV² for $c\bar{c}$) bracket the accepted value. The effective couplings respect the $\alpha_s^{c\bar{c}} > \alpha_s^{b\bar{b}}$ ordering of the running coupling; Fig. 21 shows the renormalization-group log-interpolation of Eq. (19) in the $1/\alpha_s - \ln \mu$ plane that fixes the (un-fitted) B_c coupling from the $c\bar{c}$ and $b\bar{b}$ anchors, together with the string tension and offset across sectors.

F. Conclusion

We have applied the Gaussian Expansion Method to the spectroscopy of heavy mesons, and we solved the Cornell-plus-Breit-Fermi Hamiltonian as a closed-form generalized eigenvalue problem in all four sectors. Under a statistically honest fit—residuals normalized by a theory-dominated physical uncertainty, the leptonic widths folded into the objective, and the whole result validated by leave-one-out cross-validation—we reproduce the bottomonium spectrum very well, to an RMS of 7.0 MeV ($\chi^2/\text{dof} = 0.67$), and the charmonium spectrum to 28.8 MeV, with the charm deviations sitting, as expected, in the relativistic high- L , high- n states. We reproduce the B_c and D ground states at the few-MeV and few-tens-of-MeV level respectively, and we predict a full B_c multiplet that agrees with both the literature and the HPQCD lattice B_c^* . The same wavefunctions give us $E1$ and $M1$ radiative widths within 10–25% of experiment and ground-state leptonic widths within $\sim 15\%$, while the η_c two-photon width comes out too large by $2.4\times$. We have traced this last discrepancy quantitatively to the non-relativistic over-concentration of the spin-singlet wavefunction at the origin—a short-distance effect of the leading-order formalism, and not of the variational solver itself, which passes both its virial and its anharmonic-oscillator validation tests.

1. Outlook

Several extensions follow naturally from here. First, on the dynamics: the systematic charm D -wave deficit

and the η_c two-photon excess both point to relativistic, $\mathcal{O}(v^2/c^2)$ physics, so the clearest path to quantitative annihilation widths is to add the leading relativistic kinetic correction together with the NRQCD relative-momentum operators at the annihilation vertex. A coupled-channel treatment of the open-charm thresholds [37]—which sit exactly where our charm D -wave assignments fail—would address both the $\psi(4160)$ mass and the $D\bar{D}$ widths in one stroke, replacing the tuned 3P_0 overlap with a proper momentum-space amplitude in generalized Jacobi coordinates. Second, on the unequal-mass sectors: a full treatment of the antisymmetric $\mathbf{L} \cdot \mathbf{S}_-$ spin-orbit mixing would upgrade our B_c P - and D -wave levels from the $L = 0$ -anchored predictions reported here to a complete multiplet. Third, on the statistics: we could extend the cross-validation to a fully Bayesian parameter posterior, propagating the α_s - b correlation that we presently neglect into every observable. Finally, the two-body solver generalizes—conceptually, if not trivially—to baryons and to multiquark (tetraquark, pentaquark) systems; the GEM has in fact already been carried over to heavy baryons in just this way [13], and the step calls for generalized Jacobi coordinates and the $SU(3)$ colour structure of the diquark and meson-meson configurations, together with the correct permutation symmetry of the colour-spin-space wavefunction. The source code and complete numerical output are available at github.com/LeventKaanOguz/Quarkonium-System.

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Appendix A: Detailed Derivations of the Analytic Matrix Elements

This appendix derives, from first principles, the closed-form expressions quoted in the main text. We work in the reduced-radial representation $u(r) = rR(r)$ and natural units $\hbar = c = 1$; the (unnormalized) Gaussian basis function of channel l is $u_n(r) = r^{l+1}e^{-\nu_n r^2}$.

1. The Gaussian radial moment

Every integral in the calculation is a special case of the moment of Eq. (13). Substituting $t = \nu r^2$, so that $r = (t/\nu)^{1/2}$ and $dr = \frac{1}{2}\nu^{-1/2}t^{-1/2}dt$,

$$\begin{aligned} \mathcal{I}_k(\nu) &= \int_0^\infty r^k e^{-\nu r^2} dr = \frac{1}{2\nu^{(k+1)/2}} \int_0^\infty t^{(k-1)/2} e^{-t} dt \\ &= \frac{1}{2} \frac{\Gamma(\frac{k+1}{2})}{\nu^{(k+1)/2}}, \end{aligned} \quad (\text{A1})$$

which is Eq. (13). The overlap $N_{nn'} \propto \int_0^\infty u_n u_{n'} dr = \mathcal{I}_{2l+2}(\nu_{nn'})$ of Eq. (15) follows immediately with $\nu_{nn'} = \nu_n + \nu_{n'}$, and the basis norm is $N_{nl} = [\mathcal{I}_{2l+2}(2\nu_n)]^{-1/2}$. Two moment ratios used repeatedly below follow from $\Gamma(z+1) = z\Gamma(z)$:

$$\frac{\mathcal{I}_{2l+4}(\nu)}{\mathcal{I}_{2l+2}(\nu)} = \frac{2l+3}{2\nu}, \quad \frac{\mathcal{I}_{2l+2}(\nu)}{\mathcal{I}_{2l}(\nu)} = \frac{2l+1}{2\nu}. \quad (\text{A2})$$

2. Kinetic matrix element

With the full three-dimensional basis ϕ_{nlm} of Eq. (11), the kinetic operator $T = -\nabla^2/2\mu$ is evaluated in the manifestly Hermitian gradient form, valid because the Gaussian boundary terms vanish,

$$\langle \phi_n | T | \phi_{n'} \rangle = \frac{1}{2\mu} \int \nabla \phi_n \cdot \nabla \phi_{n'} d^3r. \quad (\text{A3})$$

The angular integral is unity for like (l, m) , and with $u'_n(r) = (l+1)r^l - 2\nu_n r^{l+2}$ (times $e^{-\nu_n r^2}$) the radial part is

$$\begin{aligned} T_{nn'} &\propto \frac{1}{2\mu} \int_0^\infty u'_n u'_{n'} dr \\ &= \frac{1}{2\mu} \left[(l+1)^2 \mathcal{I}_{2l}(\nu_{nn'}) - 2(l+1)\nu_{nn'} \mathcal{I}_{2l+2}(\nu_{nn'}) \right. \\ &\quad \left. + 4\nu_n \nu_{n'} \mathcal{I}_{2l+4}(\nu_{nn'}) \right], \end{aligned} \quad (\text{A4})$$

which is exactly the three-term expression assembled by the solver. Integrating by parts once, $\int u'_n u'_{n'} = -\int u_n u''_{n'}$, and using the moment ratios (A2) collapses Eq. (A4) to the compact symmetric form $T_{nn'} = \frac{2l+3}{\mu} (\nu_n \nu_{n'} / \nu_{nn'}) N_{nn'}$ quoted in Eq. (16) [11]. The two forms are identical matrix elements; the gradient form is used numerically for its symmetry.

3. Potential and Breit–Fermi elements

The Cornell terms are single moments: the Coulomb element carries $\langle r^{-1} \rangle \propto \mathcal{I}_{2l+1}$, the linear element $\langle r \rangle \propto \mathcal{I}_{2l+3}$, and the constant $\propto \mathcal{I}_{2l+2}$, giving Eq. (16). The Breit–Fermi elements of Eq. (17) are expectation values over the converged eigenvectors. The spin-orbit and tensor pieces require $\langle r^{-3} \rangle$ and $\langle r^{-1} \rangle$, which for the integrand $r^{2l} r^{-3} = r^{2l-3}$ reduce to the *finite* moments $\mathcal{I}_{2l-1}$ and $\mathcal{I}_{2l+1}$ for every $l \geq 1$ – the $r^2 dr$ measure of the three-dimensional integral analytically tames the $1/r^3$ singularity, the key advantage over grid solvers. The smeared hyperfine element is the only one whose Gaussian width is shifted, $\nu_{nn'} \rightarrow \nu_{nn'} + \sigma^2$, a direct consequence of multiplying two Gaussians by the regulator (6).

4. Hypervirial relation for $|R(0)|^2$

The origin density of Eq. (27) follows from the $l=0$ radial Schrödinger equation $u'' = 2\mu(V-E)u$ with $u(0) = 0$ and $R(0) = u'(0)$ [5]. Forming the total derivative

$$\frac{d}{dr} [u'^2 - 2\mu(V-E)u^2] = -2\mu V'(r) u^2, \quad (\text{A5})$$

where the cross terms cancel on using the radial equation, and integrating from 0 to ∞ (both u and u' vanish at infinity, and $u(0) = 0$),

$$-u'(0)^2 = -2\mu \int_0^\infty V'(r) u(r)^2 dr \implies |R(0)|^2 = 2\mu \left\langle \frac{dV}{dr} \right\rangle. \quad (\text{A6})$$

For the Cornell potential $V' = (4\alpha_s/3)r^{-2} + b$ (the constant c and the smeared hyperfine term contribute negligibly at this order), giving Eq. (27) with $\langle r^{-2} \rangle$ evaluated analytically from $\mathcal{I}_{2l-2}(\nu_{nn'})$. Evaluating the origin density this way, rather than by summing the individually large basis amplitudes at $r=0$, is numerically stable and underlies every S -wave annihilation width.

5. S – D mixing

The tensor operator of Eq. (7) couples $L=0$ and $L=2$ for $J=1$, so the physical 1^{--} states diagonalize the 2×2 mass matrix in the $\{n^3 S_1, n'^3 D_1\}$ basis,

$$\mathcal{M} = \begin{pmatrix} M_S & V_{SD} \\ V_{SD} & M_D \end{pmatrix}, \quad V_{SD} = \sqrt{8} \frac{4\alpha_s}{3m_1 m_2} \langle r^{-3} \rangle_{SD}, \quad (\text{A7})$$

with eigenvalues $M_{\pm} = \frac{1}{2}(M_S + M_D) \pm \frac{1}{2}\sqrt{(M_S - M_D)^2 + 4V_{SD}^2}$ and the off-diagonal S_{12} factor $\sqrt{8}$ of Sec. I A 4. The lower eigenvalue is assigned to the predominantly S -wave state (e.g. $\psi(2S)$) and the upper to the predominantly D -wave state (e.g. $\psi(3770)$); the same eigenvectors mix the leptonic amplitudes through $R(0)$ and $R''(0)$.

Appendix B: Optimized Cornell Potential Parameters

The fitted parameters and their fixed inputs are listed in Table X. For $c\bar{c}$ and $b\bar{b}$ all four of (α_s, b, c, σ) are varied; the quoted 1σ errors are the square roots of the diagonal of the covariance matrix obtained from the Jacobian at the minimum. For the B_c sector α_s and σ are fixed by the renormalization-group interpolation of Eq. (19) ($\alpha_s^{bc} = 0.448 \pm 0.020$, $\sigma^{bc} = 1.923 \pm 0.231$, with uncertainties inherited from the $c\bar{c}$ and $b\bar{b}$ fits), and only b and c are tuned to the two measured masses. For the D sector only the offset c is fitted, the other three parameters being frozen to physical values. The constituent masses m_1, m_2 are inputs in every case.

TABLE X. Fixed inputs (m_1, m_2) and fitted Cornell parameters (α_s, b, c) with the hyperfine smearing scale σ . Quantities without an error are frozen/interpolated, not freely fitted (see text). The last column is the reduced χ^2 of the combined mass+width objective; “—” marks the exactly determined sectors.

Sector	α_s	b (GeV ²)	c (GeV)	σ (GeV)	m_1 (GeV)	m_2 (GeV)	χ^2/dof
Charmonium ($c\bar{c}$)	0.536(43)	0.133(11)	-0.010(60)	1.248(226)	1.500	1.500	2.53
Bottomonium ($b\bar{b}$)	0.349(9)	0.195(5)	+0.024(19)	3.103(497)	4.730	4.730	0.38
B_c ($b\bar{c}$)	0.448	0.157	+0.033	1.923	4.730	1.500	—
D ($c\bar{u}$)	0.500	0.180	-0.505	0.850	1.500	0.330	—

Appendix C: Predicted B_c Spectrum

Table XI lists the full predicted B_c ($b\bar{c}$) spectrum through the $1D$ multiplet, with the propagated model uncertainty. Only the 1^1S_0 and 2^1S_0 states are measured (and used as the fit anchors); all other entries are genuine predictions, in good agreement with the quark-model and lattice values of Refs. [9, 22, 23] (Fig. 9).

TABLE XI. Predicted B_c ($b\bar{c}$) masses (GeV) with propagated model uncertainty. The P - and D -wave triplet labels carry the unresolved $\mathbf{L} \cdot \mathbf{S}$ mixing discussed in Sec. I A 2.

State	Calc.	State	Calc.
1^1S_0	6.274(31)	1^3S_1	6.346(23)
2^1S_0	6.871(15)	2^3S_1	6.902(13)
3^1S_0	7.254(11)	3^3S_1	7.276(10)
1^1P_1	6.768(11)	1^3P_0	6.704(17)
1^3P_1	6.764(12)	1^3P_2	6.784(10)
2^1P_1	7.158(9)	2^3P_0	7.100(13)
1^1D_2	7.029(8)	1^3D_1	7.023(9)
1^3D_2	7.032(8)	1^3D_3	7.029(7)

Appendix D: GEM Wavefunction Coefficients

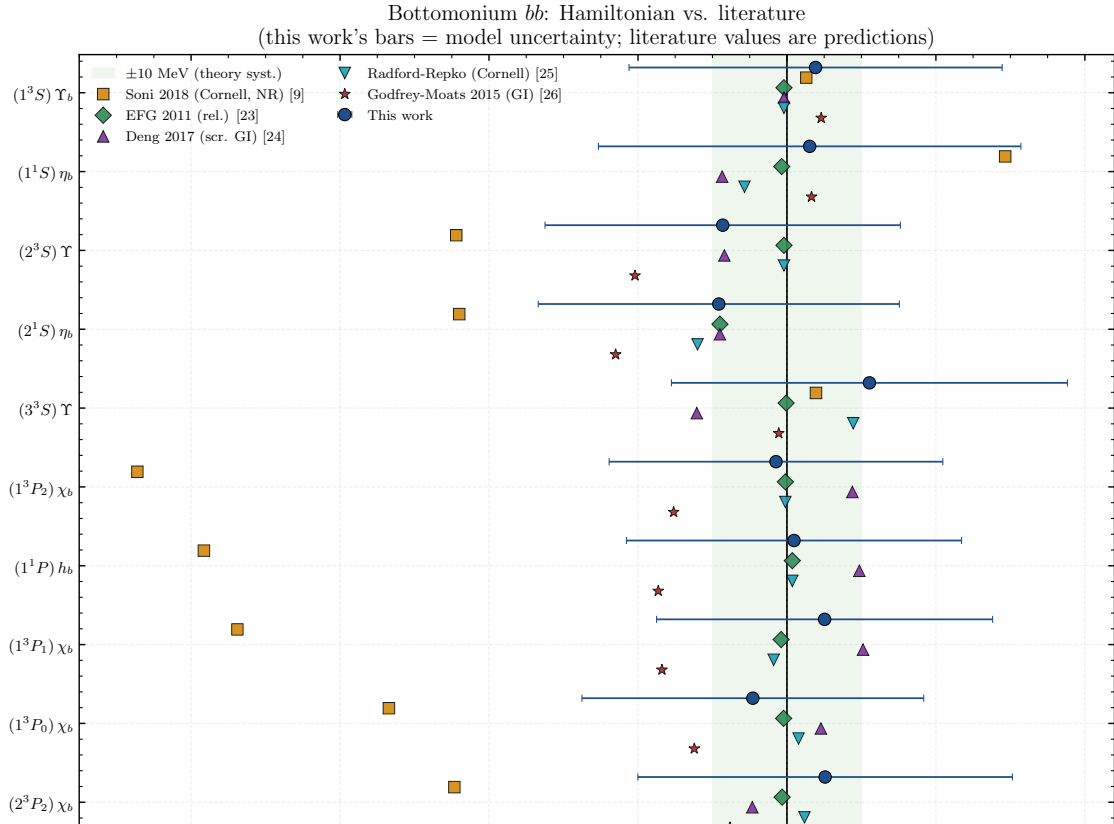
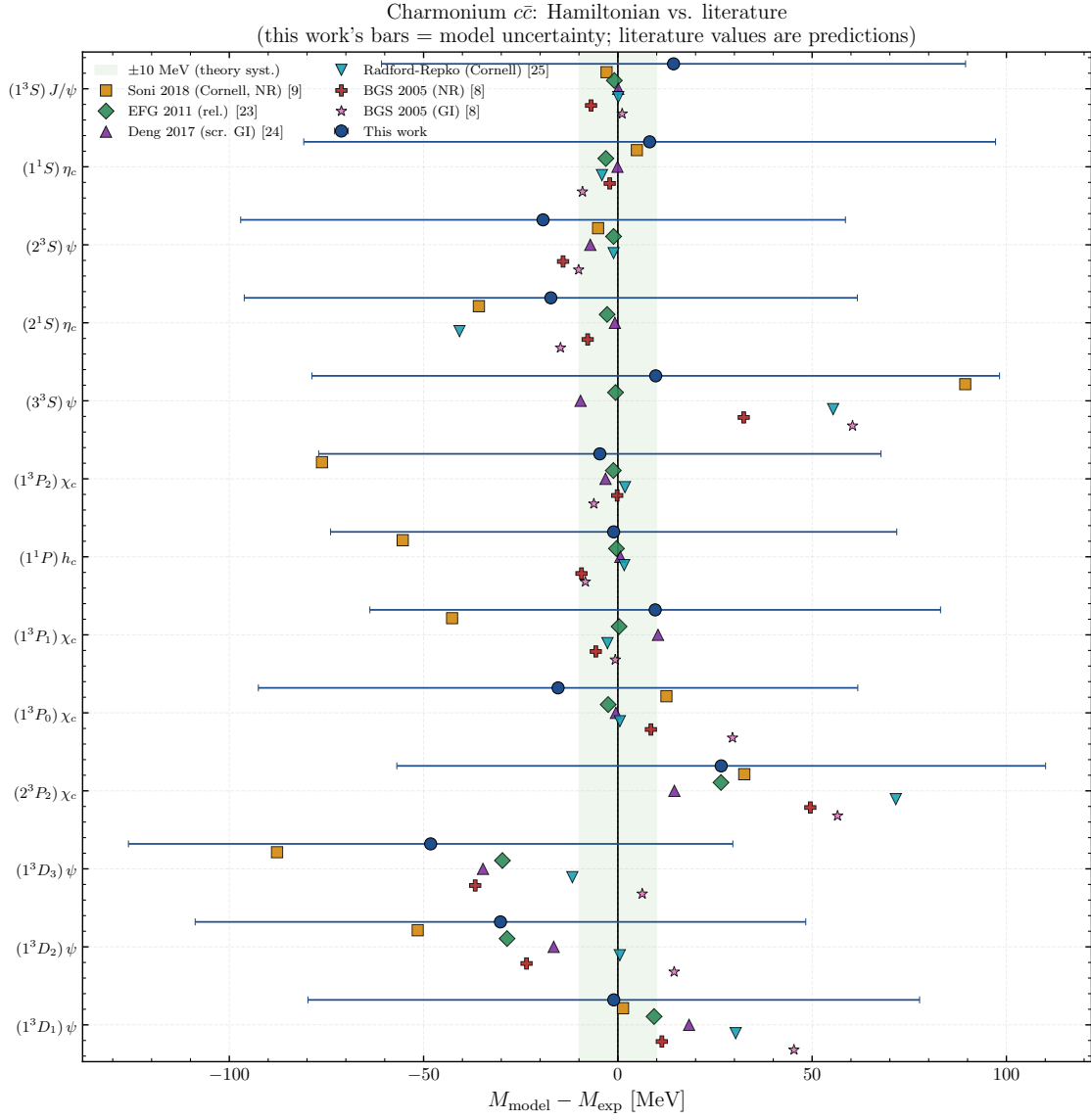
For illustration, Table XII lists the geometric Gaussian widths ν_n and the optimized expansion coefficients $C_n^{(\alpha)}$ for the S -wave bottomonium states. The radial amplitude reconstructed from the $1S$ column is the solid curve of Fig. 1. The large, oscillating coefficients of the $3S$ state reflect the partial linear dependence of the non-orthogonal Gaussian basis; this near-degeneracy is controlled, but not eliminated, by the geometric mesh and the diagonal preconditioning of Sec. II A, and is the reason the virial check (Sec. V B) is performed on every extracted state.

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TABLE XII. GEM widths ν_n and optimized expansion coefficients for the S -wave bottomonium ($b\bar{b}$) states. The heavier b quark yields a smaller effective Bohr radius, so the optimizer weights the tighter Gaussians (higher ν_n) more heavily than in the charmonium sector.

n	ν_n (GeV ²)	$C_n^{(1S)}$	$C_n^{(2S)}$	$C_n^{(3S)}$
1	400.000000	0.001220	0.000785	0.000694
2	248.675626	-0.003057	-0.001963	-0.001771
3	154.598918	0.006272	0.004020	0.003703
4	96.112457	-0.007563	-0.004819	-0.004687
5	59.752063	0.011258	0.007152	0.007250
6	37.147204	-0.008844	-0.005476	-0.006680
7	23.094011	0.015987	0.009980	0.012148
8	14.357294	-0.003739	-0.001516	-0.007113
9	8.925773	0.031585	0.019988	0.027895
10	5.549055	0.020214	0.015887	-0.002583
11	3.449787	0.063021	0.042313	0.068454
12	2.144695	0.072680	0.064523	0.015737
13	1.333333	0.146500	0.126360	0.223764
14	0.828919	0.199748	0.250742	0.168530
15	0.515330	0.290863	0.466197	1.075482
16	0.320375	0.228935	0.445871	0.002594
17	0.199174	0.054669	-1.337014	-5.842278
18	0.123824	-0.007633	-0.459843	4.884632
19	0.076980	0.004590	0.106996	-0.197441
20	0.047858	-0.002881	-0.051236	0.122201
21	0.029753	0.001739	0.026866	-0.068902
22	0.018497	-0.000951	-0.013497	0.035641
23	0.011499	0.000436	0.005869	-0.015675
24	0.007149	-0.000147	-0.001911	0.005123
25	0.004444	0.000027	0.000339	-0.000910

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Aggregate accuracy per model (this work, hatched, is fitted to these masses; others predict)

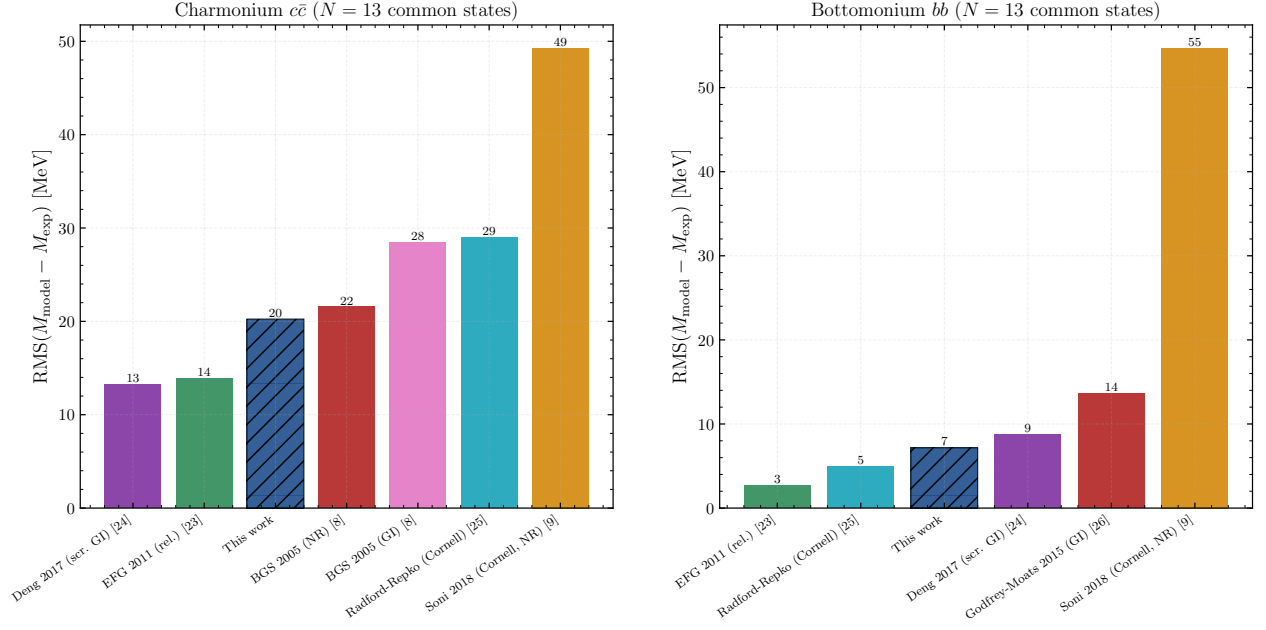


FIG. 8. Aggregate RMS of $M_{\text{model}} - M_{\text{exp}}$ over a common set of measured states. This work (hatched) is fitted to these masses; the literature models predict them, so their bar heights are the meaningful comparison.

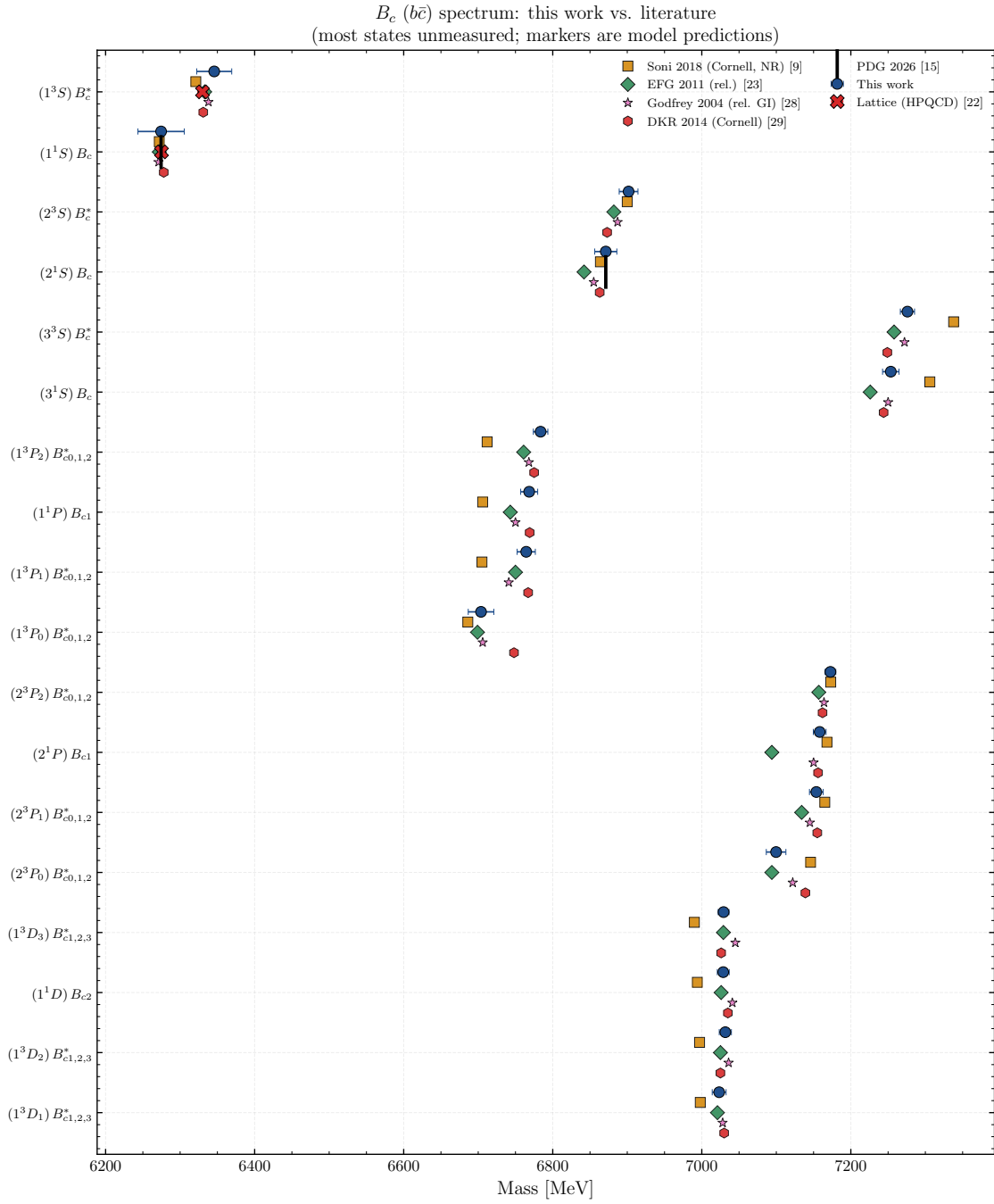


FIG. 9. Predicted B_c ($b\bar{c}$) spectrum (this work, with model uncertainty) against literature quark-model calculations [9, 23, 28, 29] and the HPQCD lattice B_c^* point [22]. The two measured states are marked.

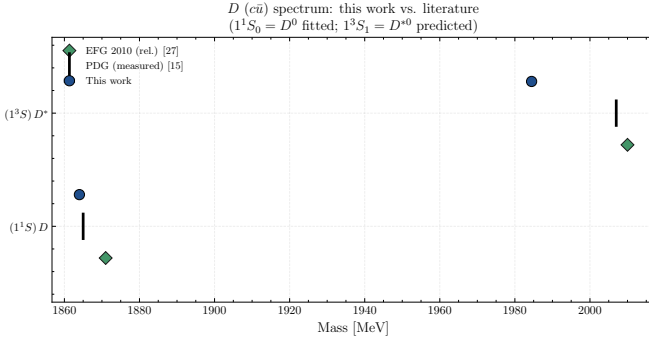


FIG. 10. $D(c\bar{u})$ ground-state doublet: this work (1^1S_0 fitted, 1^3S_1 predicted) against the relativistic quark model of Ref. [27] and the PDG-measured D^0/D^{*0} anchors.

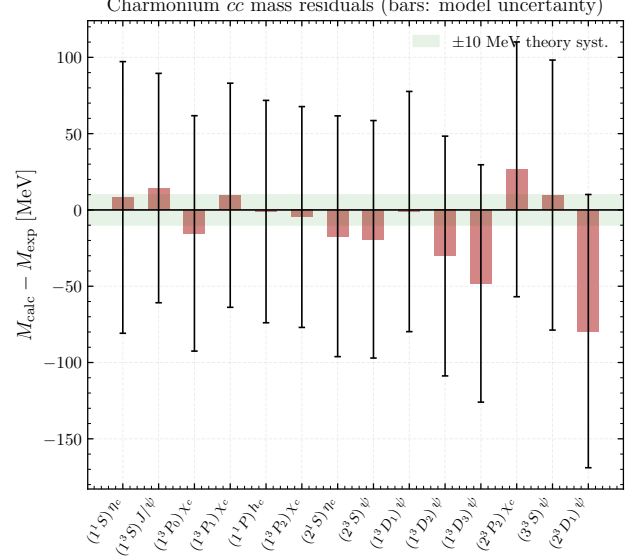
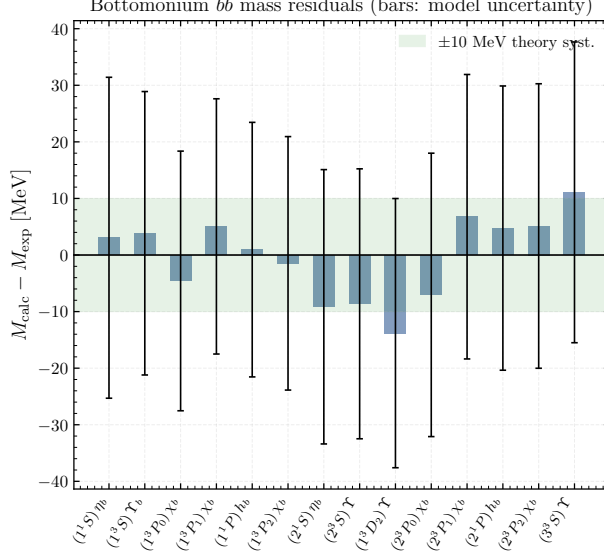


FIG. 11. Mass residuals $M_{\text{calc}} - M_{\text{exp}}$ per state for bottomonium (left) and charmonium (right), with propagated model uncertainty and the ± 10 MeV theory band.

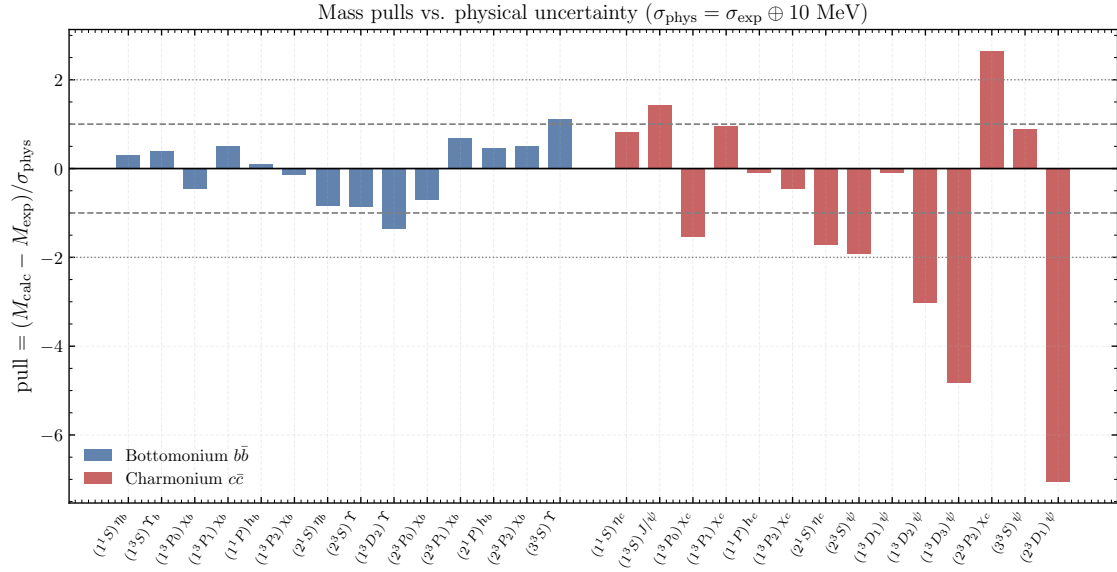


FIG. 12. Mass pulls $(M_{\text{calc}} - M_{\text{exp}}) / \sigma_{\text{phys}}$ with $\sigma_{\text{phys}} = \sigma_{\text{exp}} \oplus 10 \text{ MeV}$. Dashed/dotted lines mark $\pm 1\sigma$ and $\pm 2\sigma$.

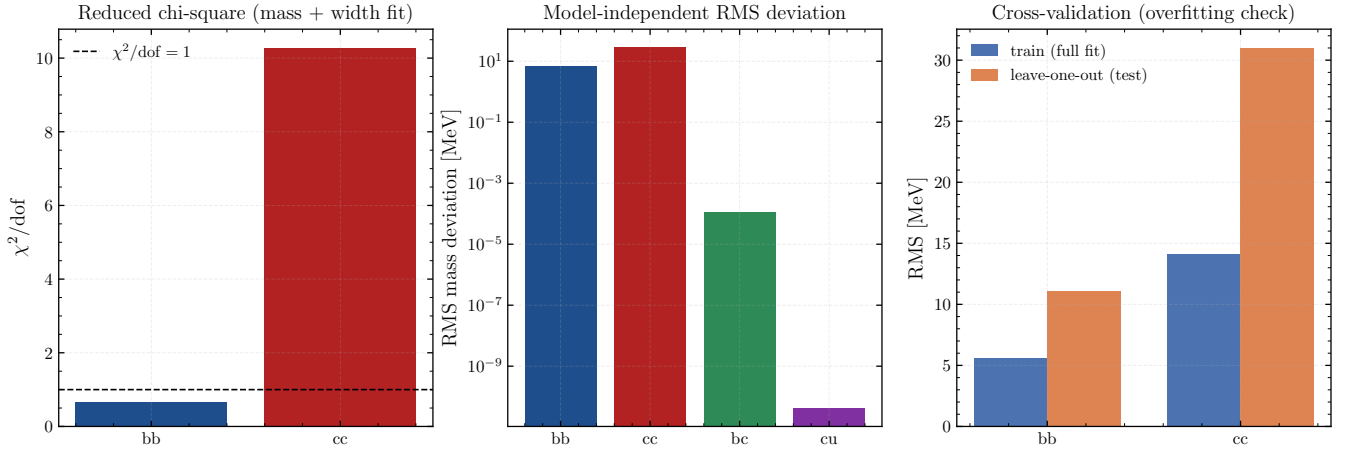


FIG. 13. Left: reduced χ^2 per sector (mass+width fit). Centre: model-independent RMS mass deviation. Right: leave-one-out cross-validation (train vs. held-out RMS), confirming the fit generalizes.

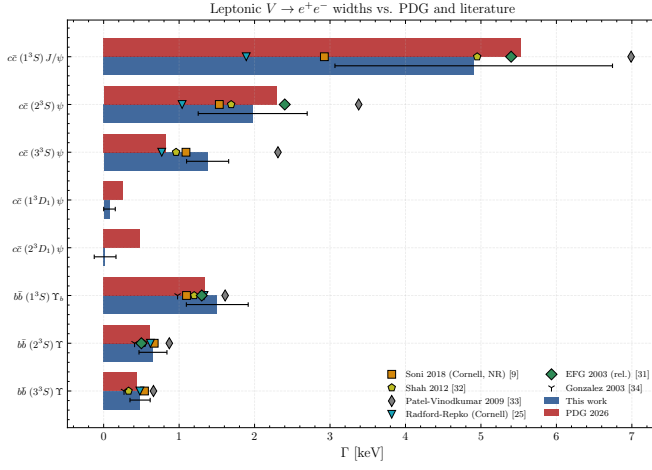


FIG. 14. Leptonic $V \rightarrow e^+e^-$ widths: this work (with propagated uncertainty) vs. PDG and the potential-model predictions of Refs. [9, 25, 31–34].

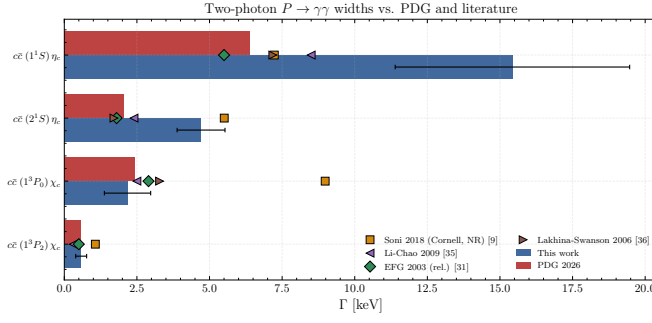


FIG. 15. Two-photon $P \rightarrow \gamma\gamma$ widths vs. PDG and the literature predictions of Refs. [7, 9, 31, 35, 36]. The S -wave η_c overestimate is the non-relativistic origin-density excess discussed in the text.

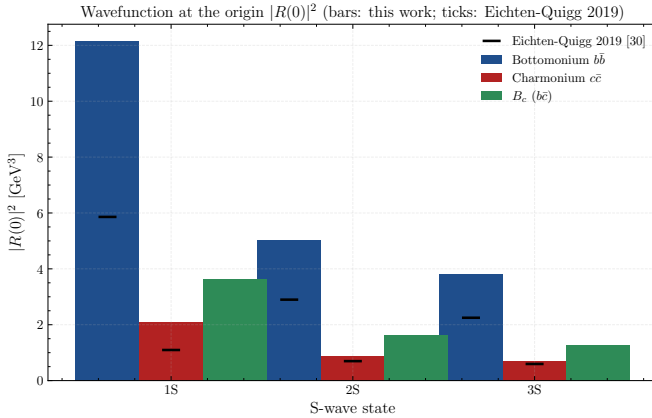


FIG. 16. Spin-singlet S -wave density at the origin $|R(0)|^2$ (bars, this work) against the Eichten–Quigg reference [30] (ticks). Our ground-state values exceed the reference by $\sim 2\times$, the source of the two-photon overestimate.

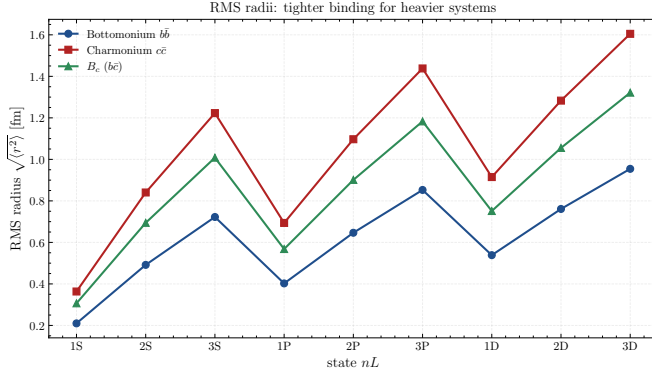


FIG. 17. RMS radii $\sqrt{\langle r^2 \rangle}$ per state and sector. The heavier $b\bar{b}$ system is the most compact; within a sector the radius grows monotonically with the radial quantum number, as expected for the Cornell potential.

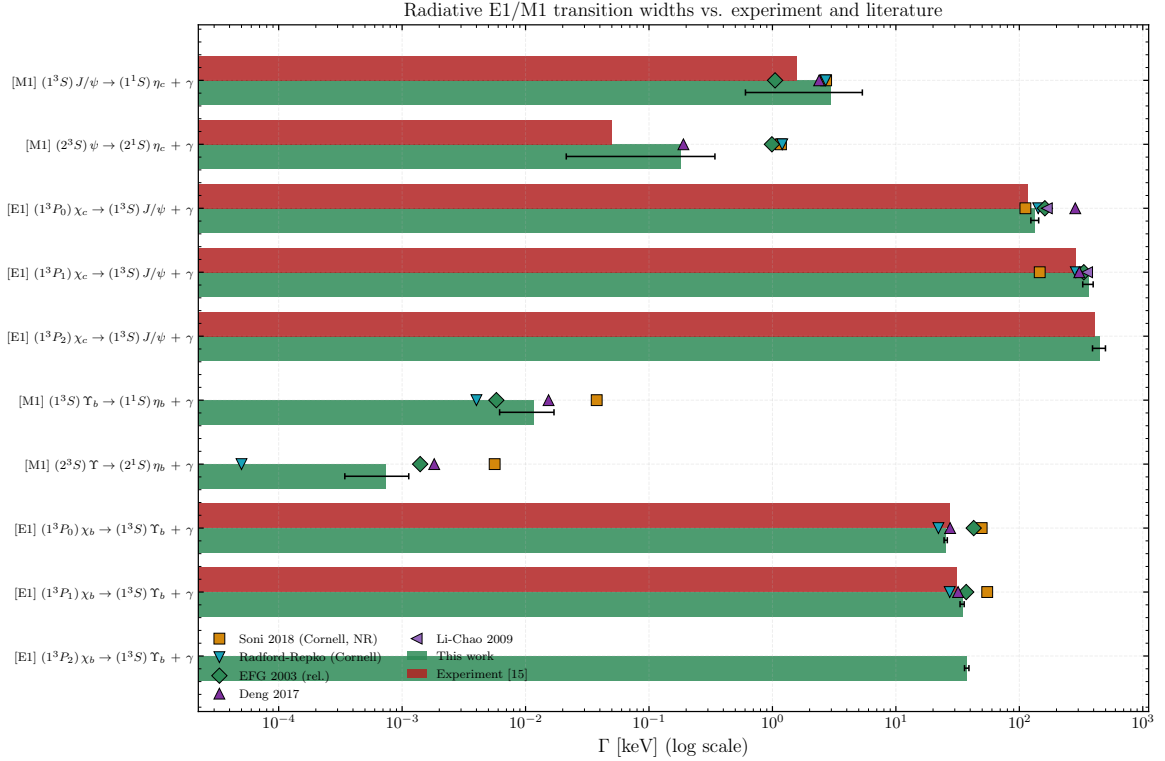


FIG. 18. Radiative $E1/M1$ transition widths (log scale) vs. experiment and the literature predictions of Refs. [9, 24, 25, 31, 35]. The $E1$ multiplets $\chi_{cJ} \rightarrow J/\psi$ and $\chi_{bJ} \rightarrow \Upsilon$ are reproduced within 10–25%.

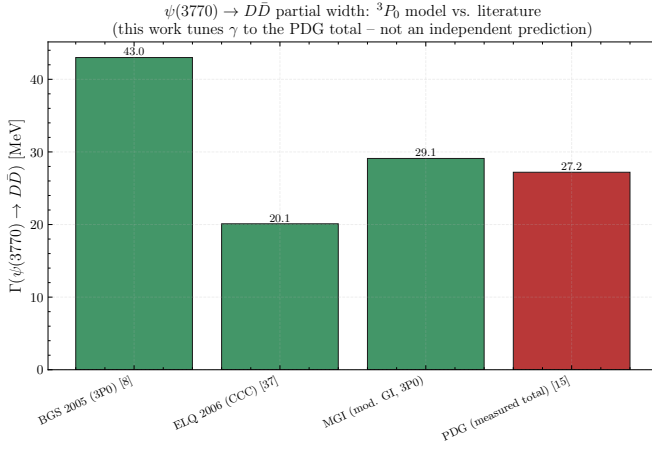


FIG. 19. $\Gamma(\psi(3770) \rightarrow D\bar{D})$ across open-charm strong-decay calculations [8, 37] against the PDG total. This work tunes γ to the PDG value (not an independent prediction).

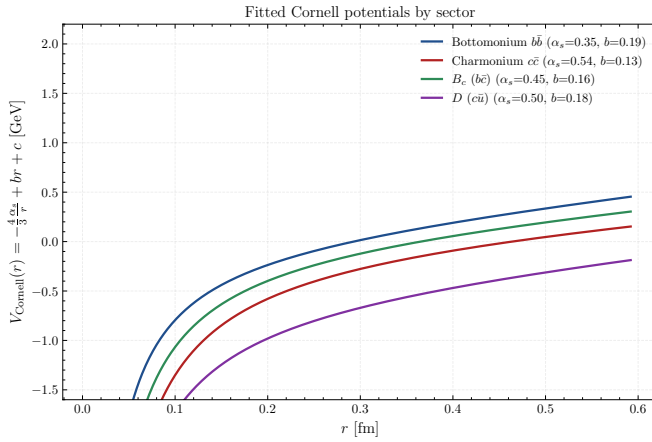


FIG. 20. The four fitted Cornell potentials $V(r) = -\frac{4}{3}\alpha_s/r + br + c$. The heavier sectors sit deeper and are probed at shorter range; the curves include the (unphysical) constant offset c that aligns the bare quark-mass sum with the physical spectrum.

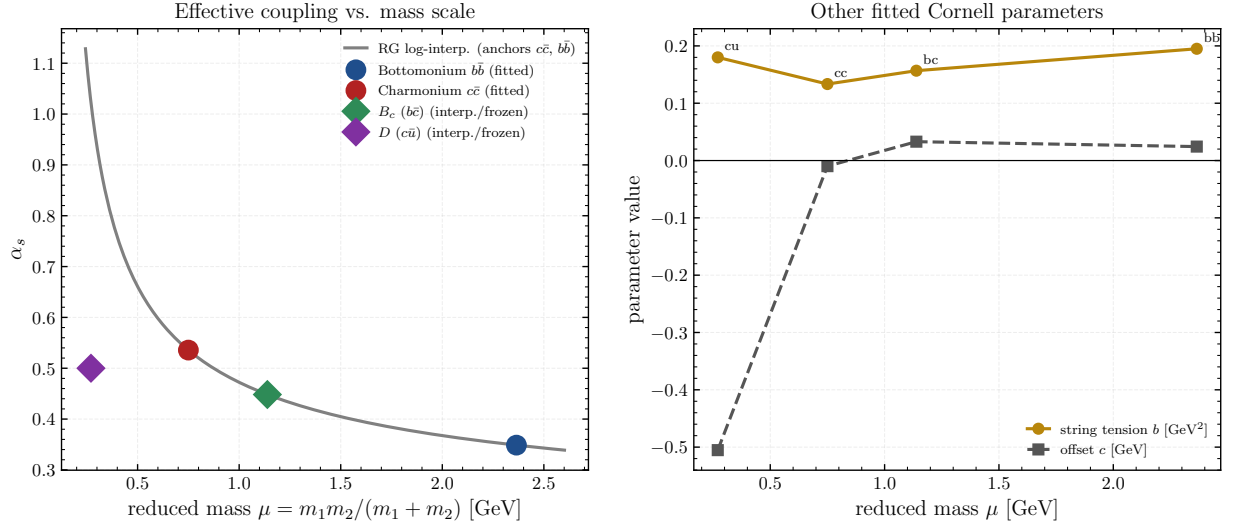


FIG. 21. Left: effective α_s vs. reduced-mass scale μ . The $c\bar{c}$ and $b\bar{b}$ fitted points (circles) anchor the renormalization-group log-interpolation (grey) of Eq. (19); the B_c coupling (diamond) is read off this curve, not fitted, and D is frozen. Right: fitted string tension b and offset c per sector.

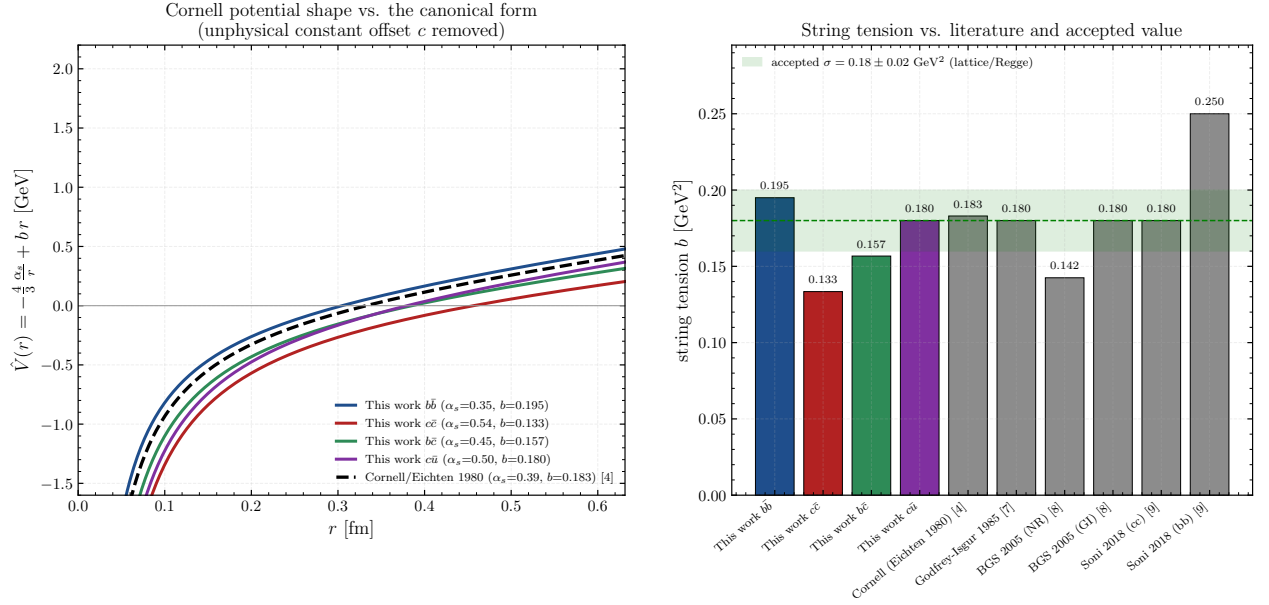


FIG. 22. Left: fitted Cornell shapes $\hat{V}(r) = -(4/3)\alpha_s/r + br$ per sector vs. the canonical Eichten 1980 form. Right: fitted string tension b vs. literature and the accepted $\sigma = 0.18 \pm 0.02$ GeV² (green band).